

Functional genomics reveals strain-specific genetic requirements conferring hypoxic growth in *Mycobacterium intracellulare*

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This investigation presents a **valuable** contribution by elucidating the genetic determinants of growth and fitness across multiple clinical strains of *Mycobacterium intracellulare*, an understudied non-tuberculous mycobacterium. Employing transposon sequencing (Tn-seq), the authors identify a core set of 131 genes essential for bacterial viability, offering a **solid** foundation for anti-mycobacterial drug discovery. However, there are minor but nonetheless significant concerns about data organization, which need to be addressed for greater scientific impact.

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Abstract

Mycobacterium intracellulare is a major etiological agent of the recently expanding *Mycobacterium avium*–*intracellulare* complex pulmonary disease (MAC-PD). Therapeutic regimens that include a combination of macrolides and antituberculous drugs have been implemented with limited success. To identify novel targets for drug development that accommodate the genomic diversity of *M. avium*–*intracellulare*, we subjected eight clinical MAC-PD isolates and the type strain ATCC13950 to genome-wide profiling to comprehensively identify universally essential functions by transposon sequencing (TnSeq). Among these strains, we identified 131 shared essential or growth-defect-associated genes by TnSeq. Unlike the type strain, the clinical strains showed increased requirements for genes involved in gluconeogenesis and the type VII secretion system under standard growth conditions, the same genes required for hypoxic pellicle-type biofilm formation in ATCC13950. Consistent

with the central role of hypoxia in the evolution of *M. intracellulare*, the clinical MAC-PD strains showed more rapid adaptation to hypoxic growth than the type strain. Importantly, the increased requirements of hypoxic fitness genes were confirmed in a mouse lung infection model. These findings confirm the concordant genetic requirements under hypoxic conditions *in vitro* and hypoxia-related conditions *in vivo*, and highlight the importance of using clinical strains and host-relevant growth conditions to identify high-value targets for drug development.

Introduction

In contrast to the ongoing decline in the rate of tuberculosis, *Mycobacterium avium*–*intracellulare* complex pulmonary disease (MAC-PD) is increasing in many parts of the world^{1,2}. The standard therapy for MAC-PD involves multidrug chemotherapy that includes clarithromycin and several antituberculous drugs. Unfortunately, treatment failure is common, resulting in relapse, disease progression, and death³. Therefore, it is critical to investigate the bacterial physiological characteristics that underlie the ability of MAC-PD strains to establish and maintain chronic drug-recalcitrant infections.

Recent advances in comparative genomics have revealed major differences in the genomic features of reference type strains and circulating pathogenic clinical isolates. Extensive genomic diversity has been recognized across clinical isolates of nontuberculous mycobacteria (NTM). *Mycobacterium avium* has been divided into four subspecies including subspecies, *avium*, *hominissuis*, *silvaticum*, and *paratuberculosis*⁴. Whereas *M. intracellulare* has been divided into two subgroups, typical *M. intracellulare* (TMI) and *M. paraintracellulare*–*M. indicus pranii* (MP-MIP), with currently two additional distinguished subspecies, *yongonense*, and *chimaera*^{5,6}. Despite the increasing number of sequenced MAC-PD strains, the molecular genetic mechanisms that underlie their virulence and pathogenesis are still poorly understood.

The burgeoning field of functional genomics has allowed the rapid assessment of molecular aspects of organisms on a genome-wide scale. Transposon (Tn) sequencing (TnSeq) is a functional genomic approach that combines saturation-level transposon-insertion mutagenesis with next-generation sequencing (NGS) to comprehensively evaluate the fitness costs associated with gene inactivation in bacteria. Over the last decade, TnSeq has been applied to numerous bacterial species using various approaches⁷. Whereas initial studies applied TnSeq to reference strains, recent approaches have included the comparative assessment of representative clinical strains to better understand the fundamental aspects of pathogen biology^{8,9}. We have previously used TnSeq to globally characterize the genes that are essential for growth versus those that are essential for hypoxic pellicle formation in the *M. intracellulare* type strain ATCC13950¹⁰. However, given the recent expansion of MAC-PD and the well-recognized high levels of genomic diversity in this complex of bacteria, data obtained from the study of a decades-old reference strain is not expected to represent the diversity of extant clinical isolates. ATCC13950 was originally isolated from the abdominal lymph node of the 34-month-old female¹¹. Even though the etiological bacterial species are the same, the clinical manifestation of infant lymphadenitis differs markedly from MAC-PD, which often occurs in middle-aged and elderly female patients with no predisposing immunological disorder. So far, there are a total of two reports on TnSeq that compare genetic requirements between clinical mycobacterial strains and the type strains in *M. tuberculosis*⁸ and *M. abscessus*⁹. They reported the diversity of genetic requirements among strains including the type strain such as *M. tuberculosis* H37Rv and *M. abscessus* ATCC19977. Therefore, it is essential to investigate a collection of strains that captures the diversity of recent clinical MAC-PD strains in order to define key molecular aspects of their pathogenesis and virulence.

In this study, we used TnSeq to analyze a set of recently isolated clinical *M. intracellulare* strains with diverse genotypes. We integrated the gene essentiality data for these strains and the type strain to identify the common essential and growth-defect-associated genes that represent the genomic diversity of this group of pathogens. We also identified the greater requirements of genes involved in gluconeogenesis, the type VII secretion system, and cysteine desulfurase in the clinical MAC-PD strains than in the type strain. Furthermore, the requirement for these genes was confirmed in a mouse lung infection model. The profiles of genetic requirements in clinical MAC-PD strains suggest the mechanism of hypoxic adaptation in NTM in terms of promising drug targets, especially for strains that cause clinical MAC-PD.

Results

Common essential and growth-defect-associated genes representing the genomic diversity of *M. intracellulare* strains

To obtain comprehensive information on the genome-wide gene essentiality of *M. intracellulare*, we included nine representative strains of *M. intracellulare* with diverse genotypes in this study (Fig. 1a, b, Supplementary Fig. 1). We used TRANSIT¹² to obtain 2–6 million reads of Tn insertions. The average number of Tn insertion reads was > 50 at Tn-inserted TA sites, which confirmed the TnSeq data (Supplementary Table 1). The numbers of essential genes in the clinical MAC-PD strains, calculated with the hidden Markov model (HMM; a transition probability model), tended to be smaller than that in ATCC13950 (Supplementary Table 2). In total, 131 genes were identified as essential or growth-defect-associated with the HMM analysis across all *M. intracellulare* strains, including the eight MAC-PD clinical strains and ATCC13950 (Fig. 2a, Supplementary Table 3). The genes identified as universal essential or growth-defect-associated were involved in fundamental functions, such as glyoxylate metabolism, purine/pyrimidine metabolism, amino acid synthesis, lipid biosynthesis, arabinogalactan/peptidoglycan biosynthesis, porphyrin metabolism, DNA replication, transcription, amino acid tRNA ligases, ribosomal proteins, general secretion proteins, components of iron–sulfur cluster assembly, some ABC transporters, and the type VII secretion system. The genes identified corresponded to the genes that encode conventional and under-development drug targets, such as those encoding gyrase, arabinosyltransferase, enoyl-(acyl-carrier-protein) reductase, alanine racemase, DNA-directed RNA polymerase, malate synthase¹³ and F₀F₁ ATP synthase (Fig. 2b). These data suggest that the set of universal essential or growth-defect-associated genes identified with the HMM analysis of *M. intracellulare* strains with diverse genotypes has potential utility as antibiotic drug targets.

By comparative genomic analysis, we have revealed 3,153 core genes among 55 *M. intracellulare* strains including those used in this study⁵ (Supplementary Fig. 1). Of these core genes, 439 genes were classified as strain-dependent essential or growth-defect-associated with the HMM analysis (Supplementary Table 4). And among 5,824 accessory genes, 222 genes were classified as essential or growth-defect-associated with the HMM analysis. The proportion of these strain-dependent and accessory essential or growth-defect-associated genes was comparable to the similar analysis of *Streptococcus pneumoniae*¹⁴. Of note, 17 (3.8%) of strain-dependent essential or growth-defect-associated genes and 14 (6.3%) of accessory essential or growth-defect-associated genes were also included in the 175 genes required for hypoxic pellicle formation in ATCC13950 reported in our previous study¹⁰ (Supplementary Table 5). Gene set enrichment analysis (GSEA)¹⁵ showed that 9.1% (16 out of 175) genes were hit as core enrichment (Supplementary Table 5). Of them, 4 genes were hit commonly as genes showing increased genetic requirements analyzed by resampling plus HMM analyses including genes of phosphoenolpyruvate carboxykinase *pckA* (OCU_RS48660), type VII secretion-associated serine protease *mycP5* (OCU_RS38275), type VII secretion protein *eccC5* (OCU_RS38345) and glycine cleavage system amino-methyltransferase *gcvT* (OCU_RS35955).

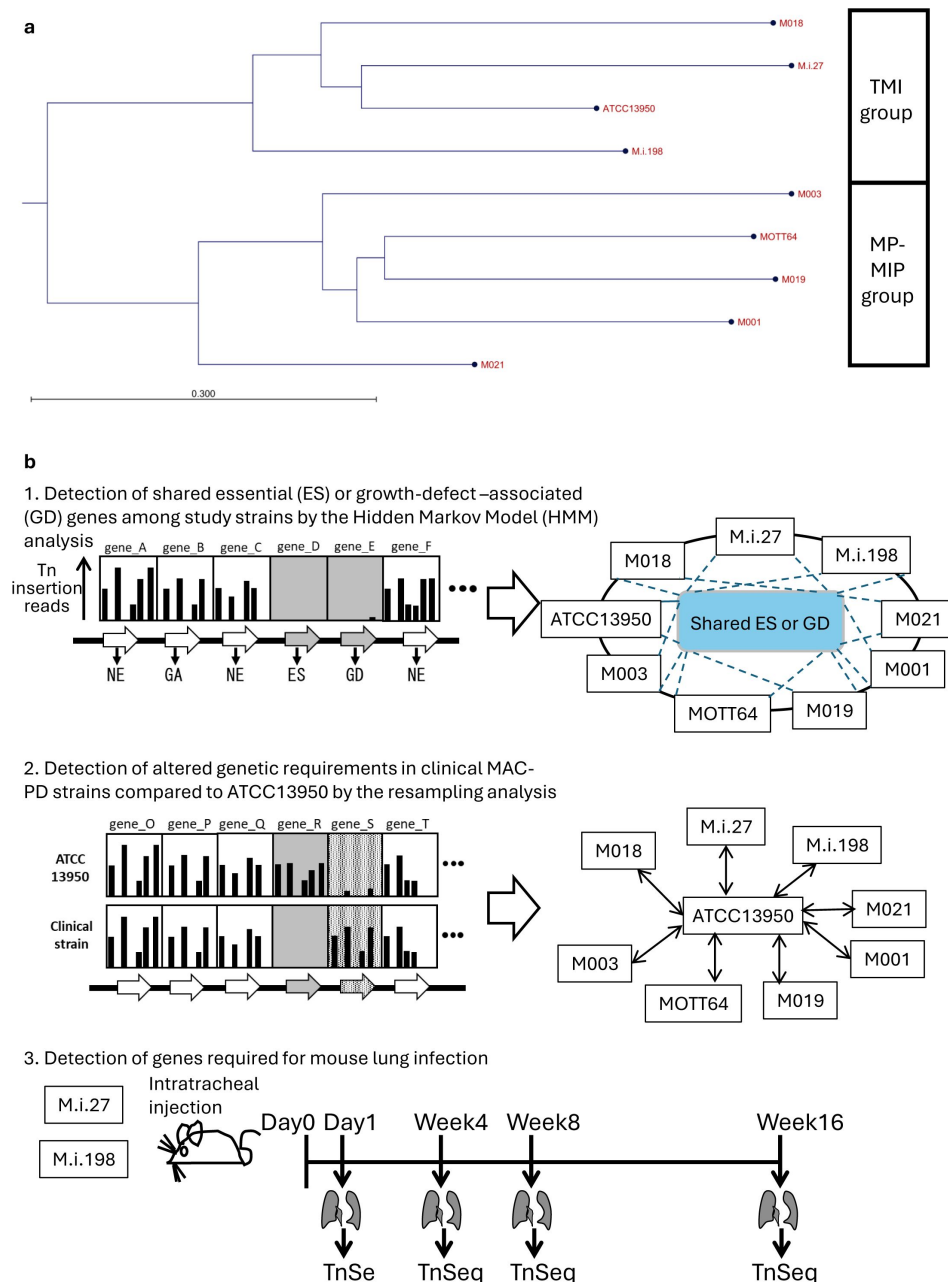


Fig. 1

Phylogenetic tree of the *M. intracellulare* strains and strategy of the TnSeq analyses in this study.

a Phylogenetic tree of the *M. intracellulare* strains used in this study. The tree was generated based on average nucleotide identity (ANI) with the neighbor-joining method. TMI: typical *M. intracellulare*; MP-MIP: *M. paraintracellulare*-*M. indicus pranii*. The detail of phylogenetic tree is shown in Supplementary Fig. 1. **b** Strategy of the procedure of TnSeq analyses.

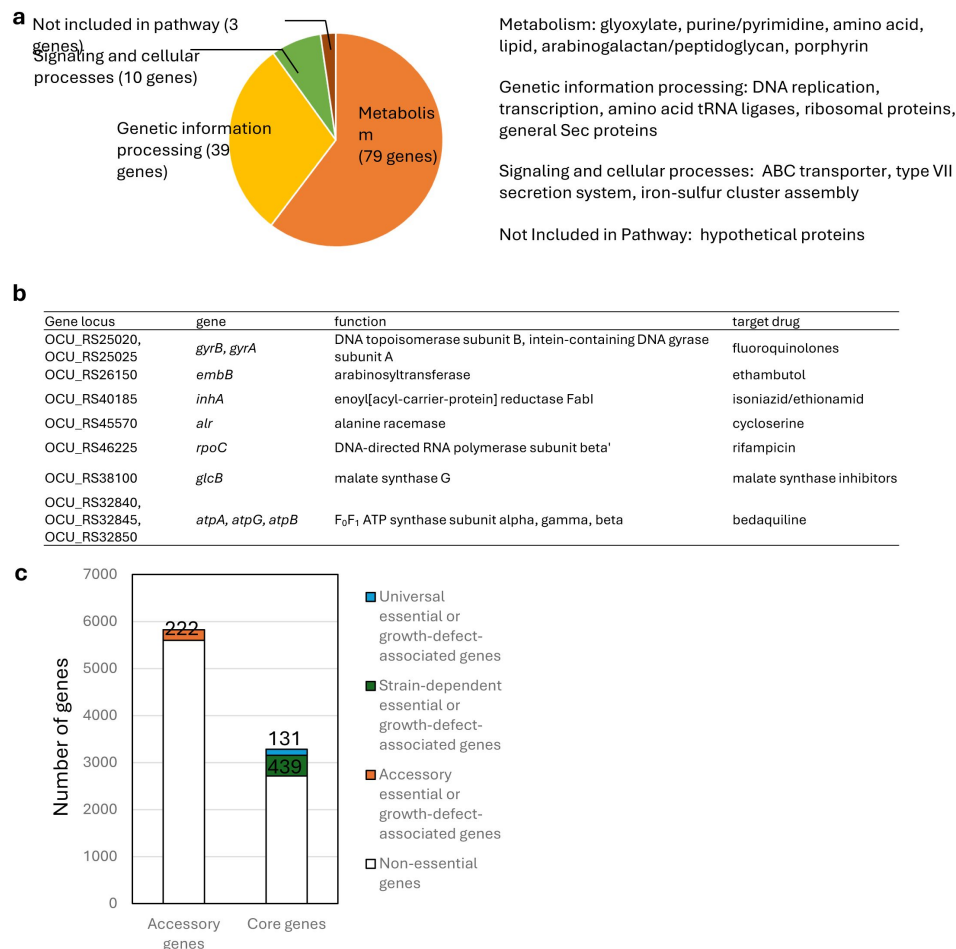


Fig. 2

Identification of the essential and growth-defect-associated genes across all nine *M. intracellulare* strains used in this study.

a Functional categories of 131 genes identified as universal essential or growth-defect-associated with an HMM analysis. The genes were categorized according to the Kyoto Encyclopedia of Genes and Genomes (KEGG) database. **b** Universal essential or growth-defect-associated genes corresponding to the genes of existing antituberculous drug targets. **c** The number of the essential or growth-defect-associated genes in the accessory genes, and the number of strain-dependent essential or growth-defect-associated genes in the core genes.

Partial overlap of the genes showing increased genetic requirements in clinical MAC-PD strains with those required for hypoxic pellicle formation in ATCC13950

The profiles of genetic requirements in each bacterial strain reflect the adaptation to the environment in which each strain lives. When the strains are placed in a special environment, they can adapt to the situation by altering the profiles of genetic requirements, resulting in the remodeling of metabolic pathways. The sharing of strain-dependent and accessory essential or growth-defect-associated genes with genes required for hypoxic pellicle formation in ATCC13950 prompts us to consider that the profiles of genetic requirements in clinical MAC-PD strains may be associated with the genes required for hypoxic pellicle formation in ATCC13950. To clarify this, we compared the genetic requirements of the clinical MAC-PD strains with that in ATCC13950 by integrating the HMM analysis with the resampling analysis, a method of calculating the number of significantly altered Tn insertion reads (a gene-based permutation model). With this integrating analysis, 121 genes in the clinical MAC-PD strains were shown to have significantly fewer Tn insertions than ATCC13950 (Fig. 3, Supplementary Tables 6, 7). Of these, nine genes were identified in 3–5 clinical MAC-PD strains. These included genes encoding phosphatases (OCU_RS30185, OCU_RS35795), cysteine desulfurase (*csd* [OCU_RS40305]), methyl transferase (OCU_RS41540), a hypothetical protein (OCU_RS47290), type VII secretion components (*mycP5* [OCU_RS38275], *eccC5* [OCU_RS38345]), a glycine cleavage system aminomethyltransferase (*gcvT* [OCU_RS35955]), and phosphoenolpyruvate carboxykinase (*pckA* [OCU_RS48660]) (Figs 3, 4). The genes involved in gluconeogenesis (such as *pckA*), the glycine cleavage system, and the type VII secretion system were also identified as required for hypoxic pellicle formation in ATCC13950 in our previous study⁹. The Tn insertion reads were significantly reduced in the pathways of gluconeogenesis, including in fructose-1,6-bisphosphatase (*glpX*), phosphoenolpyruvate carboxykinase (*pckA*), and pyruvate carboxylase (*pca*), but were significantly increased in those of glycolysis, including in pyruvate kinase (*pykA*) and pyruvate dehydrogenase (*aceE*, *lpdC*), suggesting that carbohydrate metabolism was metabolically remodeled in the clinical MAC-PD strains, which mimics the metabolism observed during hypoxic pellicle formation in ATCC13950 (Fig. 5).

On the other hand, 162 genes were detected with significantly more Tn insertions in the clinical MAC-PD strains than in ATCC13950 (Fig. 3). Of these, 12 genes were identified in more than seven strains. The increased Tn insertion reads in the genes encoding superoxide dismutase (OCU_RS25945) and pyruvate dehydrogenase (*aceE* [OCU_RS35720]) in the clinical MAC-PD strains suggest the inessentiality of the detoxification of oxygen radicals and glycolysis metabolism in these strains (Figs 3–5). These genes also included those encoding amidohydrolases (OCU_RS34400, OCU_RS34425, OCU_RS34445), some transcription factors (OCU_RS28440, OCU_RS30230), a CoA transferase (OCU_RS34390), and some predicted transporters (OCU_RS28435, OCU_RS39950, OCU_RS39955, OCU_RS44685). Furthermore, the increase in Tn insertion reads in predicted multidrug-resistant genes, such as those encoding the DrrAB-ABC transporter complex (OCU_RS39950, OCU_RS39955)¹⁶ and a putative lipoprotein LpqB (OCU_RS44685), encoded by the two-component system *mtrAB* operon^{17–18}, in the clinical MAC-PD strains were found.

In order to investigate the effect of gene duplication on the change of genetic requirements between strains, we conducted blast search for the 162 genes showing increased Tn insertion reads in the clinical MAC-PD strains compared to ATCC13950. We found that M019 has duplicate genes of OCU_RS44705 coding adenosylhomocysteinase (LOCUS_42940: *ahcY_1*, LOCUS_21000: *ahcY_2*). However, there were no duplicate genes found in the remaining 161 genes. These data suggest that gene duplication has minor effects on the change of genetic requirements between strains. Rather, sequence differences and accessory genes may play a key role in determining the difference of genetic requirements.

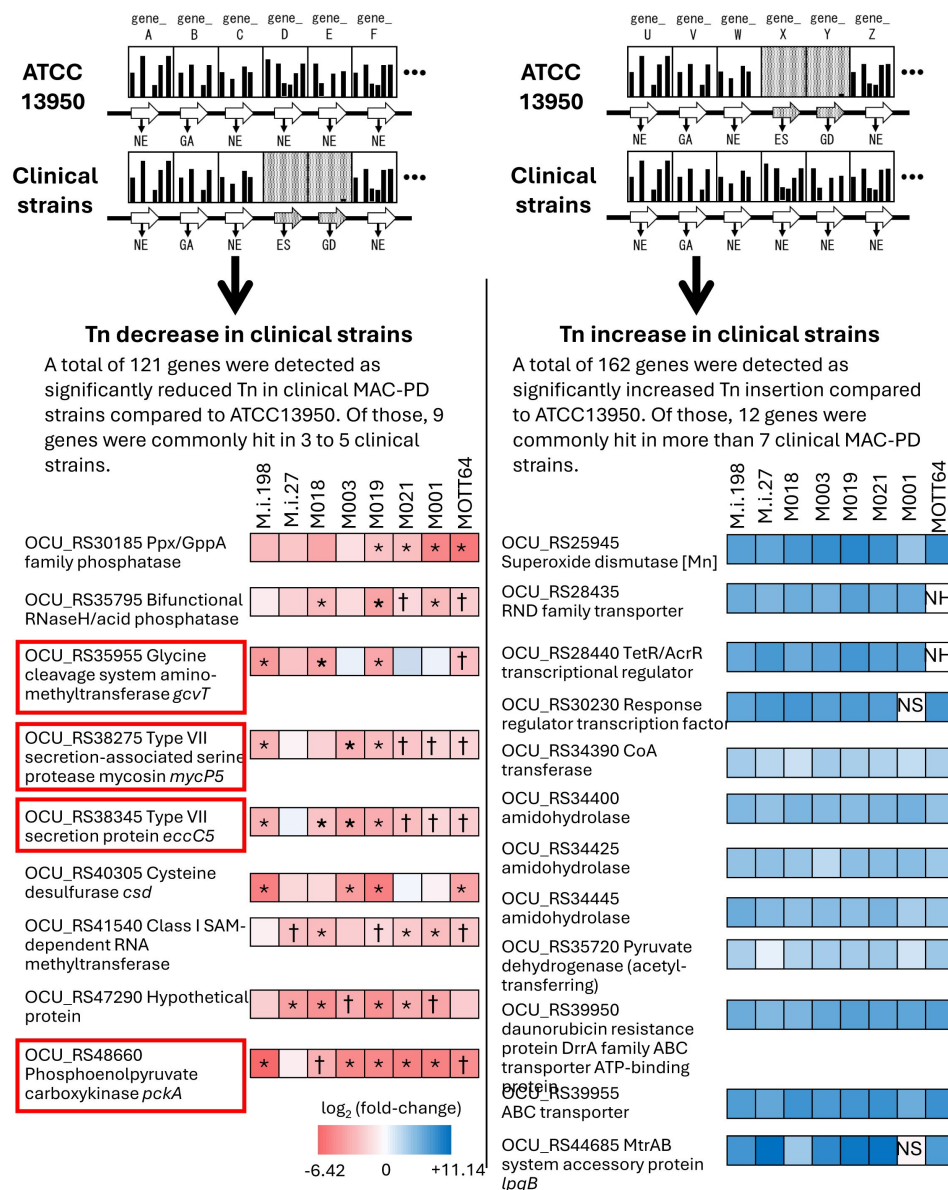


Fig. 3

Detection of genes showing increased or reduced genetic requirements in the clinical *M. intracellulare* strains.

Left panel shows the genes identified as having fewer Tn insertion reads than the type strain ATCC13950. The fold changes in the number of Tn insertion reads calculated by a resampling analysis are represented by the color scale. Red squares indicate genes required for hypoxic pellicle formation in ATCC13950. *Genes identified with a combination of resampling and HMM analyses. †Genes identified only with resampling analysis. Right panel shows the genes identified as having more Tn insertion reads than the type strain ATCC13950. NH: no homolog with ATCC13950; NS: no significant increase in Tn insertion reads by a resampling analysis (adjusted *P* < 0.05).

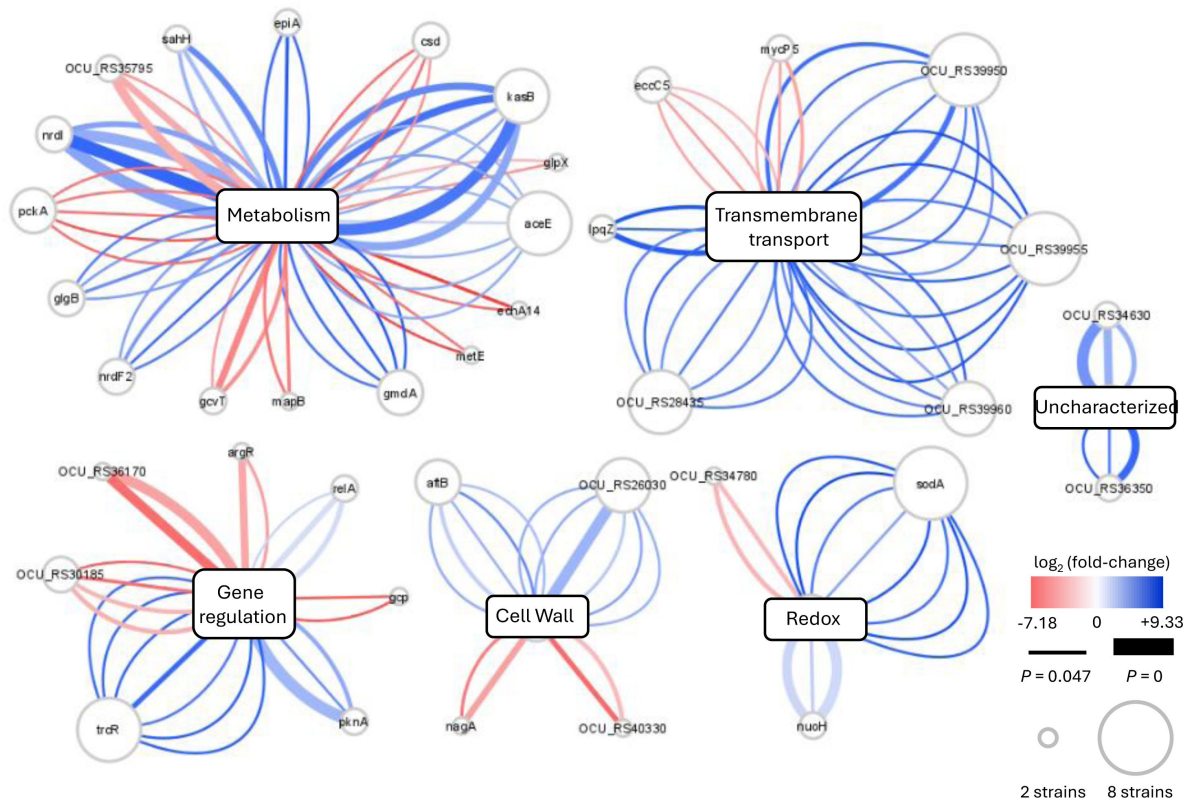


Fig. 4

Overview of the differences in genetic requirements between the clinical

M. intracellulare strains and ATCC13950, drawn with Cytoscape⁶².

Each central node represents the functional category of genes, assigned with a KEGG pathway analysis. Each peripheral node represents the genes showing significant changes in the number of Tn insertion reads. The size of each node represents the number of strains identified. Each edge represents a strain identified as showing significant changes in the number of Tn insertion reads. The thickness of each edge represents the adjusted *P* value. The color scale of each edge represents the fold change in the number of Tn insertion reads calculated by a resampling analysis. The graphical organization was referenced to the previous publication by Carey, *et al*⁸.

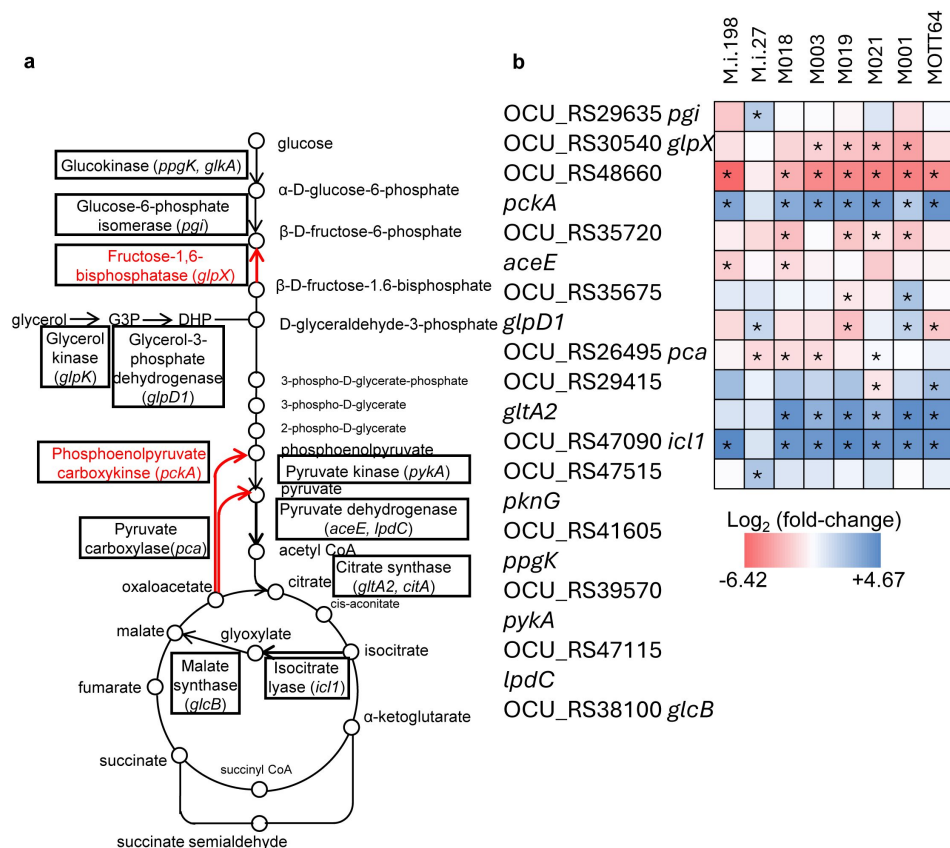


Fig. 5

Preferential genetic requirements for gluconeogenesis in the clinical *M. intracellulare* strains inferred from the TnSeq results.

a Carbohydrate pathway showing the changes in Tn insertion reads. Genes identified with the resampling analysis are framed by squares. The essentiality of gluconeogenesis-related genes (*pckA*, *glpX*) was higher and that of glycolysis-related genes (*aceE*, *lpdC*) was lower in the clinical *M. intracellulare* strains compared with those in ATCC13950. **b** Data on genetic requirements calculated with a resampling analysis in the clinical *M. intracellulare* strains compared with ATCC13950. Red indicates a reduction in Tn insertion reads in the clinical *M. intracellulare* strains. Blue indicates an increase of Tn insertion reads in the clinical *M. intracellulare* strains. Asterisks indicate statistical significance of log₂ fold changes (log₂FCs). The graphical organization was referenced to the previous publication by Carey, *et al.*⁸

Identification of genes in the clinical MAC-PD strains required for mouse lung infection

Hypoxia is the major environmental condition in natural water¹⁹, biofilms²⁰, and infections in humans and animals characterized by caseous granuloma lesions²¹. The impact of hypoxia on mycobacteria under various ecological circumstances implies that the genes required for pathogenesis of MAC-PD may be in some degrees, overlapped with the genes with increased requirements in the clinical MAC-PD strains compared to ATCC13950, and also with the genes required for hypoxic pellicle formation in ATCC13950. To identify genes required for *in vivo* infection of clinical MAC-PD strains, we performed TnSeq in mice infected intratracheally with the Tn mutant library strains of M.i.198 or M.i.27, which we had demonstrated to be virulent in mice (Fig. 1b, Supplementary Fig. 3)²². The time course of the changes in the bacterial burden showed a pattern similar to those of the wild-type strains M.i.198 and M.i.27, respectively, except that it was not possible to harvest sufficient colonies (as few as 10⁴/mouse) in the few mice infected with the M.i.27 Tn mutant strain in week 8 and week 16 (Supplementary Fig. 3b, Supplementary Table 8). With a resampling analysis, 629 and 512 genes were shown to be required for *in vivo* infection with transposon mutant library strains M.i.27 and M.i.198, respectively, at least at one time point between day 1 and week 16 of infection (Fig. 6a, Supplementary Fig. 3c, d, Supplementary Tables 9, 10). Of these genes, 72 (11.9%) and 50 (9.76%) in M.i.27 and M.i.198, respectively, were also required for hypoxic pellicle formation in ATCC13950. Forty-one genes also required for hypoxic pellicle formation in ATCC13950 were identified in both M.i.27 and M.i.198, and included genes associated with succinate production (multifunctional 2-oxoglutarate metabolism enzyme) and the glyoxylate cycle (isocitrate lyase), cysteine and methionine synthesis (thiosulfate sulfurtransferase [CysA], 5-methyltetrahydrofolate-homocysteine methyltransferase [MetH]), the serine/threonine-protein kinase PknG system, proteasome subunits (PrcA, PrcB), DNA repair UvrC, secretion protein EspR, cell-wall synthesis proteins (such as phosphatidylinositol [polyprenol-phosphate-mannose-dependent alpha-[1-2]-phosphatidylinositol pentamannoside mannosyltransferase], peptidoglycan (penicillin-binding protein A pbpA), and L,D-transpeptidase 2), putative membrane proteins and transporters (such as the MmpL family, transport accessory protein MmpS3, transport protein MmpL11, vitamin B₁₂ transport ATP-binding protein BacA, ABC transporter substrate-binding protein, and ABC transporter permease), and mammalian cell entry (Mce) proteins (OCU_RS48855–48875) (Supplementary Table 11).

At all timepoints from Day1 to Week 16, there were 74 and 99 genes were shown to be required for lung infection with M.i.27 and M.i.198, respectively. Of them, 21 (28.3%) and 12 (12.1%) genes belonged to the genes involved in hypoxic pellicle formation in the type strain. At timepoints throughout Week 4 to Week 16 except for Day1, there were 121 and 172 genes were shown to be required for lung infection with M.i.27 and M.i.198, respectively. Of them, 21 (23.1%) and 30 (18.0%) genes belonged to genes involved in hypoxic pellicle formation in the type strain. These hypoxic pellicle-associated genes detected both in M.i.27 and M.i.198 encoded methionine synthesis (MetH), acyl-CoA dehydrogenase, isocitrate lyase, MMPL family transporter (MmpL11) (from Day1 to Week 16), additionally multifunctional oxoglutarate decarboxylase/dehydrogenase, proteasome subunits (PrcA, PrcB), ABC transporter ATP-binding protein/permease, lipase chaperone (from Week 4 to Week 16) (new Supplementary Tables 12 and 13).

The genes only identified in M.i.27 that were also required for hypoxic pellicle formation in ATCC13950 included genes associated with gluconeogenesis (*pckA*, *glpX*), the glycine cleavage system (*gcvT*), the type VII secretion system (*eccD5*, *espG5*, *eccC5* [OCU_RS38280-OCU_RS38285-OCU_RS38345], *eccC3* [OCU_RS48145]), and D-inositol 3-phosphate glycosyltransferase (Supplementary Table 11). These three systems, gluconeogenesis, the glycine cleavage system, and the type VII secretion system, also showed increased genetic requirements in 3–5 clinical MAC-PD strains than in ATCC13950 (Figs 3, 6). Although M.i.27 was excluded as a relevant strain showing

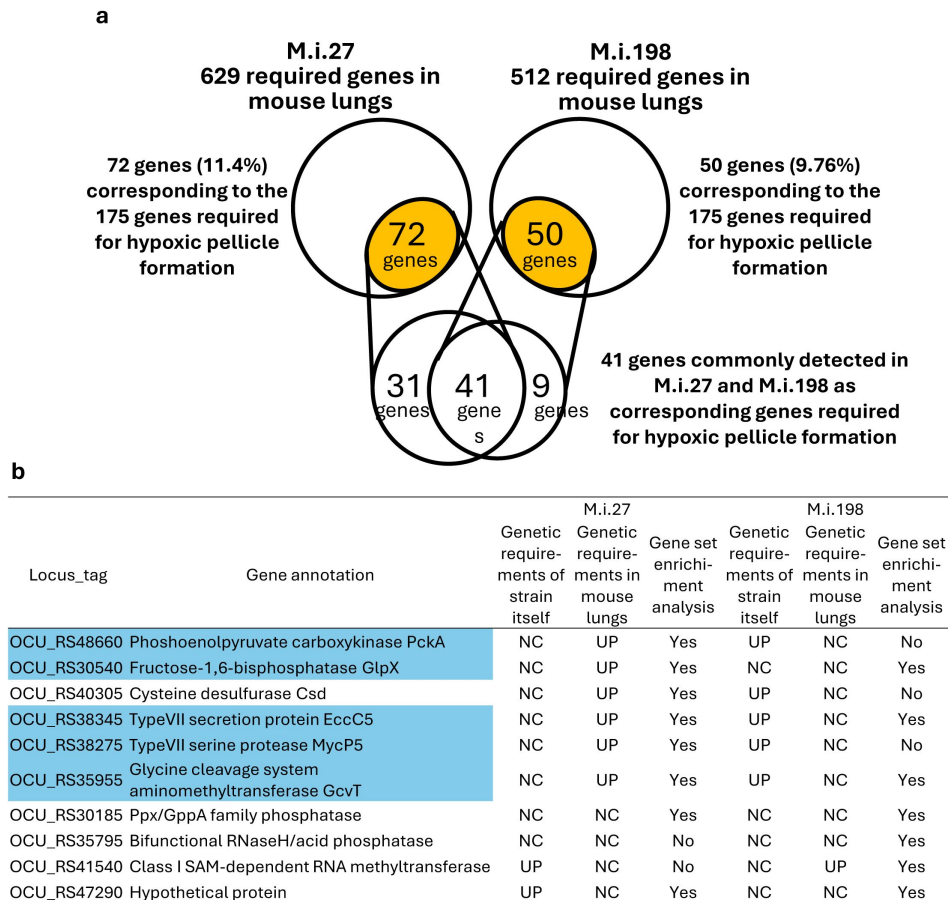


Fig. 6

Detection of genes required for mouse lung infection in the clinical *M. intracellulare* strains.

a Consistency between the genes required for infection in mouse lungs and the 175 genes required for hypoxic pellicle formation in ATCC13950. **b** Changes in the requirements for genes with increased genetic requirements observed in the clinical *M. intracellulare* strains after the infection of mouse lungs. Highlighted genes are also required for hypoxic pellicle formation by ATCC13950. Genetic requirements of the strain itself: genetic requirements compared with that of ATCC13950. Genetic requirements in mouse lungs: changes in the genetic requirements in the infected mouse lungs compared with those before infection. Gene set enrichment analysis: The genes listed as core enrichment are shown as “Yes” and the genes not listed as core enrichment are shown as “No.”

increased genetic requirements of these three metabolic systems, the increase in genetic requirements for infection *in vivo* suggests a positive relationship between hypoxic growth *in vitro* and bacterial growth *in vivo*. In contrast, M.i.198 already showed increased genetic requirements in these three metabolic systems under aerobic conditions *in vitro*, so the genes associated with these three systems were not identified as having fewer Tn insertion reads during infection *in vivo*. Genes only identified in M.i.198 and required for hypoxic pellicle formation in ATCC13950 included genes associated with succinate synthesis (2-oxoglutarate oxidoreductase subunit KorA) or encoding the HTH-type transcriptional repressor KstR and menaquinone reductase (Supplementary Table 11). As increased genetic requirements in 3–5 clinical strains (excluding M.i.198) compared with ATCC13950, an increased requirement for the RNA methyltransferase OCU_RS41540 was identified in M.i.198 during infection *in vivo*. These *in vivo* TnSeq data suggest that, despite the differences in the profiles of the genes required for *in vivo* infection between strains, increase of genetic requirements for hypoxic growth in part contribute to the pathogenesis *in vivo*.

We have performed a statistical enrichment analysis of gene sets by GSEA¹⁵ to investigate whether the genes required for growth of M.i.27 and M.i.198 in mouse lungs overlap the gene sets required for hypoxic pellicle formation in ATCC13950 as well as the gene sets showing increased genetic requirements observed in the clinical MAC-PD strains listed in Fig 3. Of these gene sets including a total of 180 genes, 54% (98 genes) and 40% (73 genes) were listed as enriched in genes required for mouse infection in M.i.27 and M.i.198, respectively ($P < 0.01$ in M.i.27 and $P = 0.016$ in M.i.198) (Supplementary Fig. 4, Supplementary Tables 14). The enriched genes both in M.i.27 and M.i.198 included genes encoding fructose 1,6-bisphosphatase (*glpX*), a type VII secretion protein (*eccC5*), glycine cleavage system aminomethyltransferase (*gcvT*), some phosphatase (OCU_RS30185) and a hypothetical protein (OCU_RS47290) (Fig. 6). The enriched genes only in M.i.27 included genes of phosphoenolpyruvate carboxylase (*pckA*), cysteine desulfurase (*csd*) and the ESX-5 type VII secretion component serine protease (*mycP5*). The enriched genes only in M.i.198 included genes encoding some phosphatase (OCU_RS35975) and class I SAM-dependent RNA methyltransferase (OCU_RS41540). Similarly, 40–50% of the genes required for ATCC13950's hypoxic pellicle formation and strain-dependent/accessory essential genes are significantly enriched individually with the genes required for *in vivo* bacterial growth (Supplementary Figs. 5, 6, Supplementary Tables 15, 16). These data suggest the sharing of the hypoxia-adaptation pathways between *in vitro* hypoxic pellicle formation and *in vivo* bacterial growth.

Preferential hypoxic adaptation of clinical MAC-PD strains evaluated with bacterial growth kinetics

The metabolic remodeling, such as the increased genetic requirements of gluconeogenesis and the type VII secretion system, and the overlap of the genes required for mouse lung infection and those required for hypoxic pellicle formation involved by these metabolic pathways suggest the preferential adaptation of the clinical MAC-PD strains to hypoxic conditions. To confirm this, we investigated the growth kinetics of each strain under aerobic and hypoxic conditions. In terms of growth kinetics, most of the clinical MAC-PD strains (except M001) entered logarithmic (log) phase earlier than ATCC13950 under 5% oxygen (Fig. 7a, 7b). However, under aerobic conditions, the clinical MAC-PD strains entered log phase at a similar time to ATCC13950. The growth rate at mid-log point (midpoint) was significantly reduced under hypoxic conditions in a total of 5 clinical MAC-PD strains (M.i.198, M.i.27, M003, M019 and M021) compared to aerobic conditions (Fig. 7a, 7c). The growth rate at midpoint under hypoxic conditions was significantly lower in these 5 clinical MAC-PD strains than in ATCC13950. By contrast, the growth rate at midpoint was significantly increased under hypoxic conditions compared to aerobic conditions in ATCC13950. And there was no significant change in the growth rate between aerobic and hypoxic conditions in the rest 3 clinical MAC-PD strains (M018, M001 and M0T64). Despite a longer lag phase compared to other strains under hypoxic conditions observed in ATCC13950 and M018, additional two strains (M005, M016) phylogenetically more closely related to ATCC13950 showed different growth

pattern under hypoxia; M005 reached midpoint later than ATCC13950, by contrast M016 reached midpoint three-quarters earlier than ATCC13950 (Supplementary Fig. 7). These data suggest that genotypes seem to have some impact on bacterial growth pattern under hypoxia; however, since there was a significant difference in the timing of hypoxic adaptation among ATCC13950 and its neighbor strains, bacterial growth pattern under hypoxia seems to be determined by multiple factors other than genotypes such as strain-specific profiles of genetic requirements and unproven regulatory systems. In summary, the data of more rapid adaptation to hypoxia followed by reduced growth rate once adapted indicate the greater metabolic advantage on continuous logarithmic bacterial growth under hypoxic conditions in the clinical MAC-PD strains than in ATCC13950.

Effects of knockdown of universal and accessory/strain-dependent essential or growth-defect-associate genes in clinical MAC-PD strains

To facilitate the functional analysis of the TnSeq-hit genes, it is necessary to construct the genetically-engineered strains. Still now, there are very few reports of gene engineering of MAC strains and none of them has been accompanied by the follow-up studies^{23–28}. Moreover, no attempts have been reported to construct the knockdown strains in MAC strains. With an intention to evaluate the effect of suppressing TnSeq-hit genes on bacterial growth, we used the pRH2052/pRH2521 clustered regularly interspaced short palindromic repeats interference (CRISPR-i) plasmids to construct knockdown strains where the expression of the target gene is suppressed^{29–31}. Of the nine *M. intracellulare* strains analyzed in this study, seven could be transformed with the vectors for single guide RNA (sgRNA) expression and dCas9 expression. However, the other two strains, M001 and MOTT64, could not be transformed with these CRISPR-i vectors. To confirm the gene essentiality detected with the HMM analysis, we evaluated the consequent growth inhibition in the knockdown strains of representative universal essential or growth-defect-associated genes, including *glcB*, *inhA*, *gyrB*, and *embB*. The knockdown strains showed growth suppression with comparative growth rates of knockdown strains to vector control strains that do not express single sgRNA reaching 10^{-1} to 10^{-4} (Fig. 8a).

As representative accessory and strain-dependent essential or growth-defect-associated genes, we evaluated the 9 genes showing reduced Tn insertion reads in the 3–5 clinical MAC-PD strains as listed in Fig. 3, as well as *glpX*, a gene of gluconeogenesis where four MAC-PD strains showed reduced Tn insertion reads in a resampling analysis (Fig. 5b). The growth was suppressed in the *pckA*-knockdown strains in M.i.198 and M003, in the *glpX*-knockdown strain in M003, and in the cysteine desulfurase gene (*csd*)-knockdown strains in M.i.198 and M003, with comparative growth rates of knockdown strains to vector control strains reaching 10^{-2} to 10^{-4} (Fig. 8b). The growth tended to be suppressed in the type VII secretion protein gene (*eccC5*)-knockdown strain in M003 and in the type VII secretion-associated serine protease mycosin gene (*mycP5*)-knockdown strain in M003, with comparative growth rates of knockdown strains to vector control strains reaching less than 10^{-1} . By contrast, the growth was not suppressed in these knockdown strains in ATCC13950, with comparative growth rates of knockdown strains to vector control strains not reaching 10^{-1} , which was consistent to the inessentiality of these genes under aerobic conditions in ATCC13950. As for M.i.198 and M003, the consistency between TnSeq data and CRISPR-i data was supported by a correlation between the growth suppression of the knockdown strains and the fitness costs evaluated by a resampling analysis (Supplementary Table 17).

In contrast to the cases of M.i.198 and M003, the growth of knockdown strains of the accessory and strain-dependent essential genes was not effectively suppressed in the other MAC-PD strains despite being positively detected in TnSeq (Fig. 8b, Supplementary Fig. 8). Although the gene expression levels were not suppressed to < 1% as reported in *M. tuberculosis* (Mtb)³², the suppression of gene expression was confirmed to approximately 20–70% in the knockdown strains

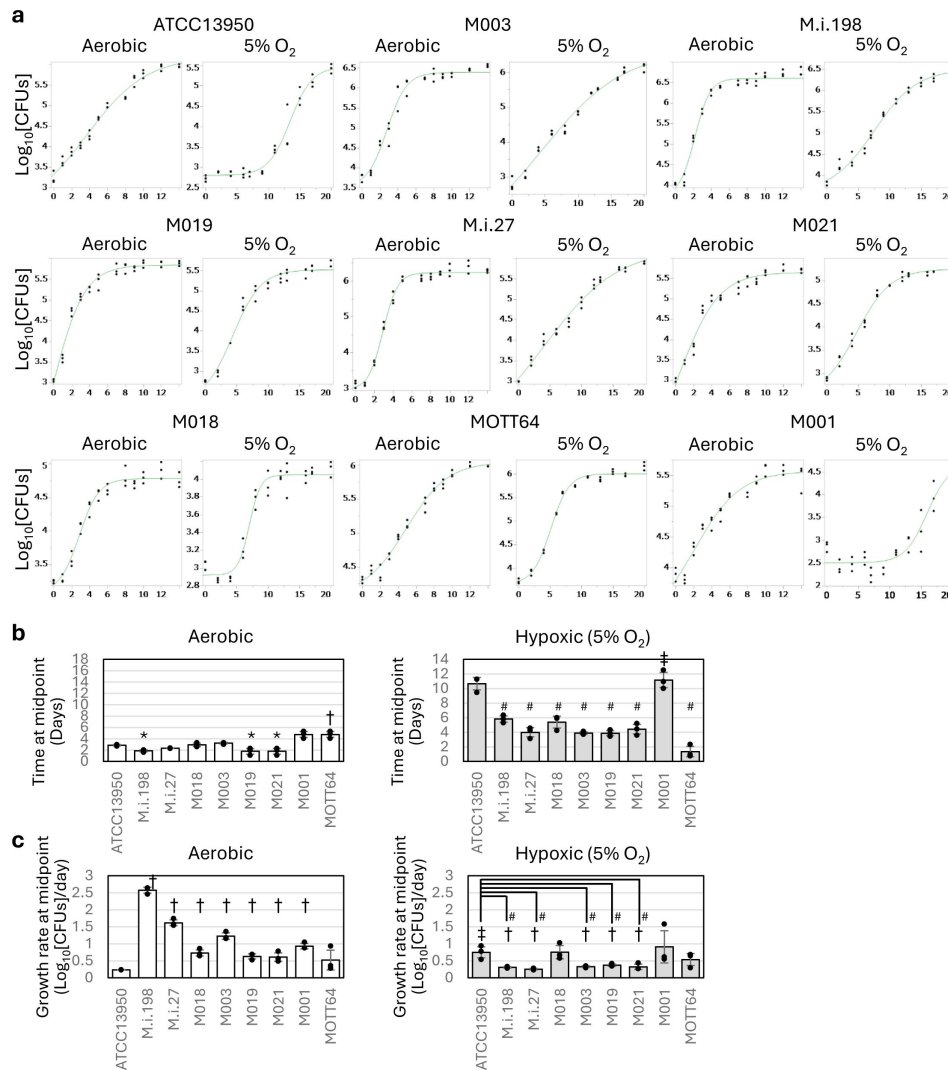


Fig. 7

Comparison of the timing of entry into logarithmic growth and logarithmic growth rate by the clinical *M. intracellulare* strains and ATCC13950.

a Representative data on the growth curves of each strain under aerobic and 5% oxygen conditions. The assay was performed three times in 96-well plates containing 250 μ l of broth medium per well, in triplicate. Data are represented as CFUs in 4 μ l sample at each timepoint. Data on the growth curves are the means of triplicate experiments. Data from one experiment representative of three independent experiments ($N = 3$) are shown. **b** Time at the inflection point (midpoint) on the sigmoid growth curve. *significantly earlier than an aerobic culture of ATCC13950; #significantly earlier than a hypoxic culture of ATCC13950; †significantly later than an aerobic culture of ATCC13950; ‡significantly later than a hypoxic culture of ATCC13950. **c** Growth rate at midpoint of the growth curve in each strain. #significantly slower than a hypoxic culture of ATCC13950; †significantly slower than an aerobic culture of the corresponding strain; ‡significantly faster than an aerobic culture of ATCC13950. Open bars: aerobic; closed bars: 5% O₂. Data are shown as the means $\pm$ SD of triplicate experiments. Data from one experiment representative of three independent experiments ($N = 3$) are shown.

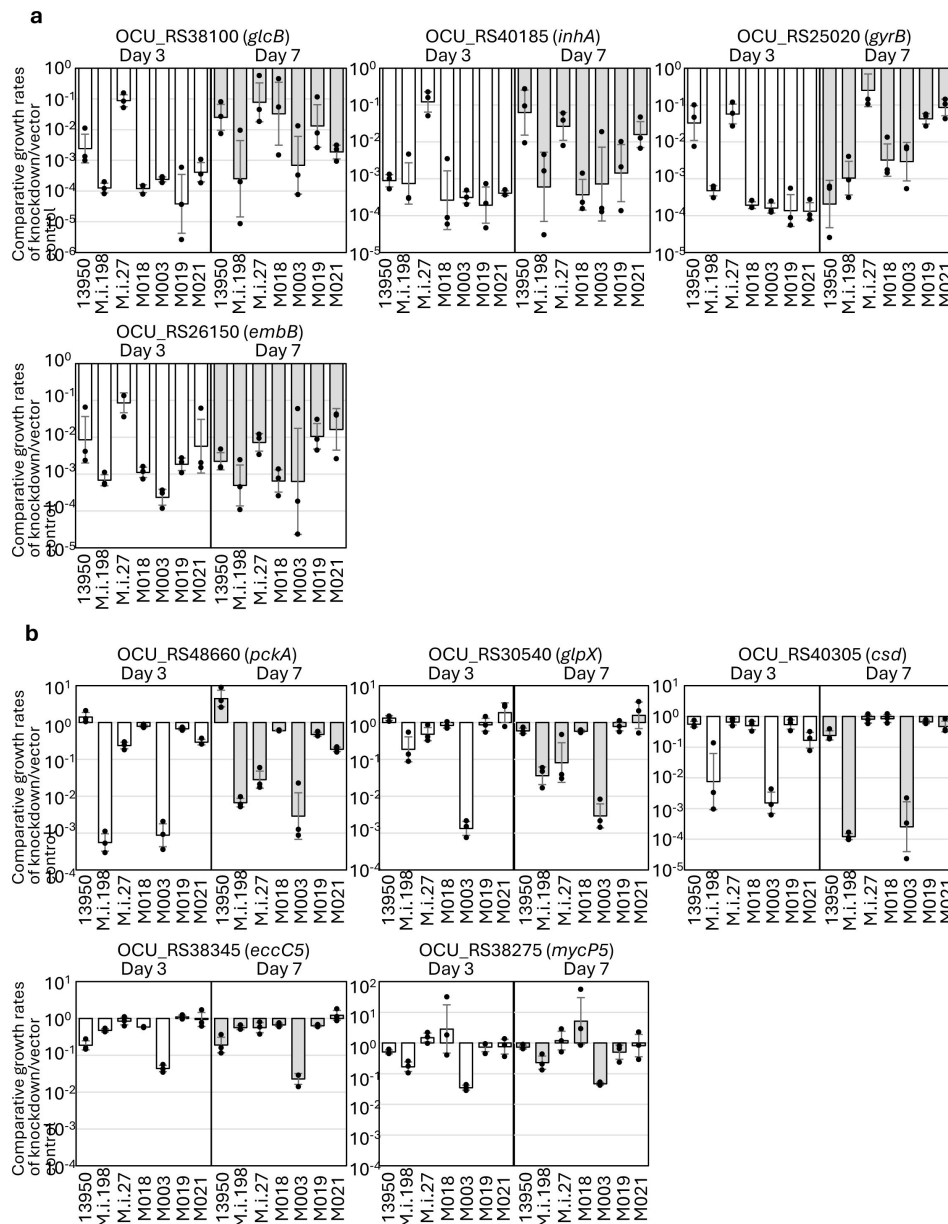


Fig. 8

Evaluation of the effect of the suppression of TnSeq-hit genes on the bacterial growth using the CRISPR-i system.

Open bar: day 3; closed bar: day 7. **a** Comparative growth rates of the knockdown strains relative to those of the vector control strains in the representative universal essential or growth-defect-associated genes: *glcB*, *inhA*, *gyrB* and *embB*. **b** Comparative growth rates of the knockdown strains relative to those of the vector control strains in the representative accessory and strain-dependent essential or growth-defect-associated genes: *pckA*, *glpX*, *csd* and ESX-5 type VII secretion components. Data are shown as the means $\pm$ SD of triplicate experiments. Data from one experiment representative of two independent experiments ($N = 2$) are shown.

of these accessory and strain-dependent essential or growth-defect-associated genes, as well as universal essential or growth-defect-associated genes such as *glcB*, *inhA*, *gyrB*, and *embB* (Supplementary Fig. 9). The possible reason may be explained by the recently revealed bypass mechanism of gene essentiality in strain-dependent/accessory essential or growth-defect-associated genes, where gene essentiality can be bypassed by several mechanisms including the composition of the accessory genome and pathway rewiring.¹⁴

To be summarized, we could validate the universal essential or growth-defect-associated genes by the CRISPR-i system in the *M. intracellulare* strains used in this study. And, although not successful with all strains, we could validate the accessory and strain-dependent essential or growth-defect-associated genes, especially genes of gluconeogenesis, type VII secretion components and iron-sulfur cluster assembly in several MAC-PD strains like M.i.198 and M003.

Discussion

In this study, using TnSeq of nine genetically diverse *M. intracellulare* strains, we have identified 131 shared essential and growth-defect-associated genes with an HMM analysis. The genes identified are promising drug targets, representing the genomic diversity of the clinical MAC-PD strains and reflecting the actual clinical situation. The strategy of narrowing the essential genes of bacteria is a useful approach for refining the database of drug targets. We have also demonstrated an overview of the differences in genetic requirements for hypoxic growth between the ATCC type strain and the clinical *M. intracellulare* strains responsible for MAC-PD, with the evidence of earlier entry to log phase followed by slower growth rate observed in most of these clinical strains compared to ATCC13950. This is the first study to demonstrate the positive relationship between the profiles of increased genetic requirements for hypoxic metabolic remodeling and the phenotypes advantageous for bacterial survival under hypoxic conditions.

In this study, 121 genes were detected as significantly reduced Tn insertion in the clinical MAC-PD strains compared to ATCC13950, which was higher than the case of *Mtb* showing 10–50 genes with altered Tn insertion reads detected by a resampling analysis⁸. This may reflect the genomic diversity of *M. intracellulare* compared with that of *Mtb*. The proportion of accessory genes is larger in NTM than in *Mtb*³³. In a clinical trial of phage therapy for *M. abscessus* infection, the number of accessory genes was shown to decrease with the loss of virulence in *M. abscessus* during the course of therapy³⁴, suggesting some role for accessory genes for its clinical pathogenesis. Our data of the overlap of the accessory essential or growth-defect-associated genes such as *pckA* and *mycP5* (Supplementary Table 6) with genes required for mouse lung infection (Fig. 6b, Supplementary Fig. 3) supports the idea that accessory genes (non-core genes) may encode functions required for bacterial survival, especially under nonoptimal conditions, such as hypoxia and during the infection of human and animal bodies.

Our TnSeq data are consistent with metabolomic data that have shown the importance of the gluconeogenic carbon flow of TCA cycle intermediates for bacterial growth under hypoxic conditions in terms of the essentiality of *PckA* and the reverse TCA cycle.^{35–38} Gluconeogenesis is essential for bacterial growth when fatty acids are used as a major nutrient.³⁷ *PckA* catalyzes the conversion of the metabolites derived from the TCA cycle to the metabolites of gluconeogenesis.³⁶ The reaction catalyzed by *PckA* is essential for bacterial growth when the available nutrients are predominantly fatty acids.^{39,40} The main nutrients for mycobacteria inside hypoxic granulomas in an infected human or mouse body are lipids, such as fatty acids and cholesterol, rather than carbohydrates.^{37,41} In the present study, *M. intracellulare* grew under 5% oxygen, whereas *Mtb* stops growing under 1% oxygen and enters a dormant state.³⁶ The depletion of phosphoenolpyruvate in the *pckA* deletion mutant of *Mtb* reduced the phosphoenolpyruvate–carbon flux essential for bacterial growth and drug

sensitivity³⁶. The present study supports the idea that PckA is essential for the microaerobic growth of mycobacteria. Our TnSeq data imply the preferential adaptation of the clinical MAC-PD strains for circumstances *in vivo*, which are characterized by hypoxia.

On the other hand, such metabolic shift towards gluconeogenesis and reverse TCA cycle can also be observed under reactive nitrogen species and *in vivo* infection as well. This suggests that the metabolic shift observed under hypoxia is a general stress response rather than being specific to hypoxia due to the nature of the provided/available carbon source⁴². Several factors other than hypoxia can be estimated to be related to the change of the genetic requirements in the clinical MAC-PD strains such as host-related factors and general stress conditions (i.e. nutrition of abundant fatty acids and less carbohydrates, low pH circumstances)^{37,39–41}. Except for our data of the profile of genes required for hypoxic pellicle formation in *M. intracellulare*¹⁰, no fundamental genome-wide data are available on the change of the genetic requirements in response to each stress factor in non-tuberculous mycobacteria. Further studies are needed to discriminate which factors are specifically involved in the increased requirements of each gene. Although of interest, it is beyond the scope of the current study.

About 10% of the genes required for mouse infection are also required for hypoxic pellicle formation. The genes showing increased genetic requirements in the clinical MAC-PD strains, such as *pckA* and genes encoding the ESX-5 type VII secretion system, including *eccC5* and *mycP5*, also showed increased genetic requirements in mouse infections. In a TnSeq study of mice infected intravenously with *M. avium*, the genes of the ESX-5 type VII secretion system were required for infection of the liver and spleen⁴³. The ESX-5 type VII secretion system is involved in the transport of fatty acids and the virulence of mycobacteria^{44,45}. These similarities in the profiles of genetic requirements prompt us to propose some role of biofilm formation in mycobacterial pathogenesis in animals and humans. In the present study, we shed light on the impact of hypoxia on biofilm formation, the acquisition of virulence, and the *in vivo* pathogenesis of NTM in terms of global genetic requirements. Although the morphologies of the pellicle and macroscopic lesions (including cavitation and nodular bronchiectasis) are not similar, monitoring the expression of the genes identified in this study may provide clues to the mechanism of biofilm formation *in vivo*.

In this study, we revealed the pattern of hypoxic adaptation characteristic to the clinical MAC-PD strains by the growth curve analysis: an early entry to log phase followed by a reduced growth rate (**Fig. 7**). This pattern implicates the continuous impact on the infected hosts during logarithmic bacterial growth, which may be involved in the persistent and steadily progressive illness for years in humans³. The advantage of slow growth in mycobacteria may be inferred by the role of a histone-like nucleoid protein, mycobacterial DNA-binding protein 1 (MDP1), also designated HupB^{46–48}. The expression of MDP1 is enhanced in the stationary phase, resulting in reduced growth speed and long-term survival with increased resistance to reactive oxygen species (ROS)^{46–48}. The reduced growth rate after entry to log phase may allow long-term survival of mycobacteria under ROS-rich circumstances inside host cells. The pattern of hypoxic adaptation as observed in the clinical MAC-PD strains may enhance pathogenesis by extending the period of the repeated process of bacterial invasion and replication in infected macrophages.

Contrary to our expectation, not all clinical strains showed rapid adaptation from aerobic to hypoxic conditions relative to ATCC13950. These data suggest that some, but not a major proportion of, clinical *M. intracellulare* strains do not have an advantageous phenotype for continuous logarithmic growth under hypoxia. In *Mtb*, the adaptation to hypoxia is considered to be necessary for clinical pathogenesis because the conditions *in vivo*, such as inside macrophages or granulomas, are hypoxic^{42,49}. However, our strain M001 entered log phase later than ATCC13950 under 5% oxygen. Our inability to construct knockdown strains in M001 prevented us from clarifying its clinical pathogenesis without preferential hypoxic adaptation. However, because we have shown that most of the *M. intracellulare* strains with different genotypes showed

preferential adaptation to hypoxia in this study, our results largely reflect the clinical situation of the etiological agents of MAC-PD. A possible strategy for identifying factors that confer clinical pathogenesis will be to focus on the major clinical strains that show preferential hypoxic adaptation.

The data of validation of the universal essential or growth-defect associated genes by the CRISPR-i system was overall acceptable in the clinically MAC-PD strains whose knockdown strains could be constructed (**Fig. 8a**). However, we have encountered the difficulties of validating the TnSeq-hit genes by CRISPR-i experiment, especially accessory and strain-dependent essential or growth-defect-associated genes in the MAC-PD strains other than M.i.198 or M003 (**Fig. 8b**, Supplementary Fig. 8). We found the lower efficiency of the suppression of gene expression by CRISPR-i in *M. intracellulare* than in *Mtb* that is reported to be suppressed to 6% under 10 ng/ml anhydrotetracycline (aTc) and by < 1% under 100 ng/ml aTc in the knockdown strains, compared to strains expressing scrambled single guide RNAs (sgRNA)³². However, because the levels of knockdown efficiency were comparable between strain-dependent/accessory essential genes and universally essential genes (Supplementary Fig. 9), the low efficiency of knockdown does not seem to explain the reason of the discrepancy between TnSeq and CRISPR-i results. Although it is difficult to find the clear definitive reason for this phenomenon only from the current study, the possible reason may be speculated by the bypass mechanism of gene essentiality which is characteristically observed in strain-dependent/accessory essential or growth-defect-associated genes. According to the publication by Rosconi¹⁴ reporting the ‘forced-evolution experiments’ of 36 clinical *Streptococcus pneumoniae* strains, gene essentiality can be bypassed by several mechanisms including the composition of the accessory genome and pathway rewiring. They recovered successfully knockout mutants from transformation experiments in strain-dependent/accessory essential genes such as cytidine monophosphate kinase, a folate pathway enzyme formate tetrahydrofolate ligase and an undecaprenyl phosphate-biosynthesis pathway enzyme farnesyl-diphosphate synthase. The bypassing of gene essentiality could be suggested by observing suppressor mutations and synthetic lethality in knockout strains. By contrast, universal essential genes were reported to fulfill the three categories including high levels of conservation within and often across species, limited genetic diversity, and high and stable expression levels. Consequently, universal essential genes were estimated to be rigid, largely immutable key components to an organism’s survival. The knockout recovery of the universal essential genes was reported to fail as shown by no colonies or only appearance of merodiploids. Taking into consideration such bypass mechanism of gene essentiality, the inconsistent effect of gene silencing of strain-dependent/accessory essential genes on bacterial growth may reflect some kinds of pathway rewiring that helps the bacteria grow under suppression of the target gene expression. Nevertheless, our study has opened the avenue for functional analysis of NTM that leads to the development of novel drugs targeting universal and accessory/strain-dependent essential metabolic pathways, especially by making use of M.i.198 and M003 as optimal NTM strains for gene manipulation.

As in the *glpX*-knockdown strains, *csd* knockdown caused growth suppression in several MAC-PD strains. This type II Csd is involved in the biosynthesis of the iron–sulfur clusters of the SUF machinery. Compared with the type I cysteine desulfurase IscS, the SUF machinery is active under oxidative-stress and iron-limiting conditions⁵⁰. Representatives of iron–sulfur cluster proteins include components of the electron-transport system, such as NADH dehydrogenase I (Nuo) and succinate dehydrogenase (Sdh), which are components of respiratory complexes I and II, respectively. Under hypoxic conditions, the non-proton-translocating type II NADH dehydrogenase Ndh is upregulated, and cytochrome *bd* oxidase (CydAB) and the proton-pumping type I NADH dehydrogenase (Nuo) are downregulated⁴². In the reverse TCA cycle on the same reaction pathway as Sdh, fumarate reductase (Fnr) is also an iron–sulfur cluster protein that produces succinate, and is a proposed alternative electron acceptor under hypoxic conditions^{38,51}. The

increased essentiality of the *csd* gene observed in the clinical MAC-PD strains seems to reflect the switching of the electron-transport system to less-energy-efficient respiration to support preferential hypoxic survival, including *in vivo*.

We tried to use inhibitors of phosphoenolpyruvate carboxykinase to confirm the increased essentiality of the *pckA* gene. However, we detected no positive effect because millimolar 3-mercaptopycolinic acid is required to show the effect of inhibition, even in eukaryotic cells, which have no cell wall⁵². Moreover, 3-alkyl-1,8-dibenzylxanthine⁵³ also only effectively inhibits PckA in eukaryotic cells. However, PckA has been proposed as a drug target for *Mtb*⁵⁴, so our TnSeq data seem to be consistent with studies directed towards antimycobacterial drug discovery.

We found the difference in capacity of transformation between strains belonging to the same genomic *M. paraintracellulare*-*M. indicus pranii* (MP-MIP) subgroup.

Although the direct mechanism determining the competency for foreign DNA has not been elucidated in *M. intracellulare* and other NTM species, several studies on general bacteria suggest the difficulties of introducing foreign DNA into clinical strains compared to the laboratory strains. As suggested in *Staphylococcus aureus*⁵⁵, some clinical strains develop elimination systems of foreign nucleic acids such as a type III-like restriction endonuclease. As suggested in gram-negative bacteria⁵⁶, there may be some difference in cell surface structures between strains, resulting in the necessity of polymyxin B nonapeptide targeting cell membrane for transforming clinical strains. The efficiency of eliminating foreign DNA may be attributed to various kinds of strain-specific factors including restriction endonuclease, natural CRISPR-interference system and cell wall structures rather than a simple genotype factor.

There may be a claim that the saturation of Tn insertion in most of each replicate were 50-60% in our libraries. However, our method of constructing Tn mutant libraries is in accordance with the previous studies of “high-density” transposon libraries⁹ because the saturation of the Tn mutant libraries by combining replicates are 62-79% as follows: ATCC13950: 67.6%, M001: 72.9%, M003: 63.0%, M018: 62.4%, M019: 74.5%, M.i.27: 76.6%, M.i.198: 68.0%, MOTT64: 77.6%, M021: 79.9%. That is, we predicted gene essentiality from the Tn mutant libraries with 62-79% saturation in each strain. However, there is non-permissive sequence pattern in mariner transposon mutagenesis and using more than 10 replicates of Tn mutant libraries is reported to be a more accurate method^{57,58}. The issue of high probability of judging nonessential region as mistakenly as “essential” are mainly found in small ORFs (2-9 TA sites) rather than larger ORFs (TA sites > = 10)⁵⁷. There are 4136 (6.43% of total ORFs) nonpermissive TA sites in ATCC13950, which is less than in *M. tuberculosis* H37Rv (9% of total ORFs)⁵⁷ and in *M. abscessus* ATCC19977 (8.1% of total ORFs)⁵⁸. As for small ORFs (2-9 TA sites), there are nonpermissive TA sites in 41 genes (ORFs) of common essential or growth-defect-associated (1.35% of a total of 3021 larger ORFs in ATCC13950). With respect to the covering of non-permissive sites, our TnSeq data may need to be improved to classify quite accurately the gene essentiality particularly on the nonstructural small regulatory genes including small RNAs, although the study by DeJesus shows the comparable number of essential genes in large genes possessing more than 10 TA sites between 2 and 14 TnSeq datasets, most of which seem to be structural genes⁵⁷.

In summary, we have identified 131 universal essential and growth-defect-associated genes that cover the genomic diversity of *M. intracellulare* strains responsible for MAC-PD, which narrowed down the drug targets against MAC-PD. Furthermore, we have demonstrated preferential adaptation to hypoxia in clinical NTM strains compared with the type strain in terms of the increased genetic requirements for hypoxic growth, such as those encoding phosphoenolpyruvate carboxykinase, the ESX-5 type VII secretion system, and cysteine desulfurase. These genes were shown to have a significant impact on infection *in vivo* by increasing their genetic requirements.

Our data highlight the importance of using clinical strains and hypoxic conditions in experimental research, and provide fundamental resources for developing novel drugs directed towards nontuberculous mycobacterial strains that cause severe clinical diseases.

Methods

Bacterial strains and phages used to prepare Tn mutants

A total of nine strains were used in this study, including seven clinical strains isolated from patients with MAC-PD in Japan (M.i.198, M.i.27, M018, M003, M019, M021, and M001)²², *M. intracellulare* genovar *paraintracellulare* type strain MOTT64 isolated from MAC-PD in South Korea⁵⁹, and the *M. intracellulare* type strain ATCC13950 (Fig. 1). The Tn mutant libraries were constructed using glucose as a carbon source under aerobic conditions according to the transduction method used for ATCC13950¹⁰, except that a higher concentration of kanamycin (100 µg/ml) was used to select the transposon mutants than was used for ATCC13950 (25 µg/ml). In brief, a hundred milliliter of the bacteria were cultured aerobically in Middlebrook 7H9 medium (Difco Laboratories, Detroit, MI, USA) containing 0.2% glycerol, 0.1% Tween 80 (MP Biochemicals, Illkrich, France), and 10% Middlebrook albumin–dextrose–catalase (7H9/ADC/Tween 80) until mid-log phase (optical density at a wavelength of 600 nm [OD₆₀₀] of approximately 0.4–0.6). After harvesting the bacteria, 10 mL of high-titer mycobacteriophage phAE180⁶⁰ was transduced into the bacteria 1-day at 37°C for inserting Tn and plated on Middlebrook 7H10 solid medium supplemented with 10% oleic acid–albumin–dextrose–catalase (Becton Dickinson and Co., Sparks, MD) (7H10/OADC) agar plates containing kanamycin. After 2 or 3 week-cultivation at 37°C aerobically, the colonies were harvested *en masse* and used as Tn mutant libraries. The kanamycin-resistant colonies ($> 1 \times 10^5$) were harvested and evenly resuspended in 7H9/ADC medium containing 20% glycerol, aliquoted, and stored at –80 °C until further use.

Next-generation sequencing and TnSeq analysis

DNA libraries were prepared as reported previously⁶¹. The TnSeq libraries were sequenced with the NextSeq 500 System, 150-bp single-end run (Illumina, San Diego, CA, USA). The sequence reads were analyzed as previously described¹⁰. Genetic requirements were calculated, and gene essentiality was predicted by resampling and HMM analyses using TRANSIT, respectively¹². First, an HMM analysis was conducted to calculate the fluctuation of the number of Tn insertion reads individually for each strain. HMM method can categorize the gene essentiality throughout the genome including “Essential”, “Growth-defect”, “Non-essential” and “Growth-advantage”. “Essential” genes are defined as no insertions in all or most of their TA sites. “Non-essential” genes are defined as regions that have usual read counts.

“Growth-defect” genes are defined as regions that have unusually low read counts. “Growth-advantage” genes are defined as regions that have unusually high low read counts. Universal essential and growth-defect-associated genes were identified as the genes classified as “Essential” and “Growth-defect” with an HMM analysis in all 9 subject strains (Supplementary Table 3). Next, a resampling analysis was conducted to calculate the difference of the number of Tn insertion reads between each clinical strain and ATCC13950 (Supplementary Table 7). By comparing the data obtained by HMM and resampling analyses, we summarized the difference of genetic requirements between the clinical MAC-PD strains and ATCC13950. To compare the genetic requirements of the clinical MAC-PD strains and ATCC13950, the annotation for the clinical MAC-PD strains was adapted from that of ATCC13950 by adjusting the START and END coordinates of each open reading frame (ORF) in the clinical MAC-PD strains according to their alignment with the corresponding ORFs of ATCC13950. On TRANSIT software, we set the replicate option as Sum to combine read counts. And we used Beta-Geometric correction (BGC) to normalize the datasets to

fit an “ideal” geometric distribution with a variable probability parameter ρ ¹². The function of TnSeq-hit genes was categorized according to the Kyoto Encyclopedia of Genes and Genomes (KEGG) database.

Comparison of growth curves under aerobic and hypoxic conditions

First, we precultured the bacteria in 7H9/ADC broth until the bacteria reached early log-phase (OD₆₀₀ approximately 0.1–0.3). Next, the bacterial cultures were diluted with new 7H9/ADC broth (final OD₆₀₀ 0.003) for the aerobic and hypoxic growth assay.

Bacterial cells were grown statically in 96-well plates containing 250 µl of 7H9/ADC broth in each well under aerobic conditions or 5% oxygen at 37 °C with humidification (APM-30D incubator, Astec, Tokyo, Japan). The culture was sampled every 1–3 days. At sampling, the culture was mixed by pipetting 50 times. An aliquot (4 µl) of the culture and its diluents were streaked onto 7H10/OADC agar and the CFUs counted.

The growth curve was fitted to the logistic 4P model with the JMP software (SAS Institute Inc., Cary, NC, USA) to calculate the inflection point (midpoint) and growth rate at midpoint in the log growth phase. Entry into the log phase was evaluated from the timing of the inflection point on the growth curve.

Animals

Female C57BL/6Jcl mice were purchased from CLEA Japan (Tokyo, Japan). Six-week-old mice were used for the infection experiment (Supplementary Fig. 3). All the mice were kept under specific-pathogen-free conditions in the animal facility of Niigata University Graduate School of Medicine, according to the institutional guidelines for animal experiments. The mice were housed (four per cage) under a 12-h/12-h light/dark cycle. Food and water were available *ad libitum*. Kanamycin was added to the drinking water to a final concentration of 100 µg/ml to maintain the mice’s resistance to the Tn mutant strains. Euthanasia was conducted with an intraperitoneal injection of a combined anesthetic (15 µg of medetomidine, 80 µg of midazolam, and 100 µg of butorphanol), followed by cervical dislocation.

Mouse TnSeq experiment

The protocol was based on our previous mouse lung infection experiment²². Twenty mice each were used for the Tn mutant strains M.i.198 and M.i.27. The mice were injected intratracheally with 3.25×10^8 and 5.93×10^7 CFUs of the Tn mutant strains, respectively. The infected lungs were expected to be harvested from five mice at day 1, week 4, week 8, and week 16 after infection. However, the lungs were harvested from only four mice infected with the M.i.27 Tn mutant strains in week 8 because we experienced difficulties with sampling, and the lungs were harvested from only three, one, and two mice infected with M.i.198 Tn mutant strains in week 8, week 12, and week 16, respectively, because morbidity was high in the mice infected with M.i.198. The harvested lungs were homogenized in 4.5 ml of distilled water with the gentleMACS™ Tissue Dissociator (Miltenyi Biotec, Bergish Gladbach, Germany) and the homogenates were plated on 7H10/OADC agar plates and cultured for 3 weeks. The colonies were harvested *en masse* for TnSeq.

Gene set enrichment analysis

To investigate the overlap of the gene sets required for growth in mouse lungs with those required for hypoxic pellicle formation observed in ATCC13950 and those showing increased genetic requirements observed in the clinical MAC-PD strains, a statistical enrichment of gene sets was

performed by GSEA 4.3.2.¹⁵ Parameters set for GSEA were as follows: permutations = 1,000, permutation type: gene_set, enrichment statistic: weighted, metric for ranking genes: Signal2Noise, max size: 500, min size: 15.

¹Construction and growth assay of knockdown strains

Knockdown strains were constructed with the CRISPR-dCas9-mediated RNA interference system, as reported previously.²⁹ Gene-specific 20-nt sequences were selected as protospacers for incorporation into the sgRNAs (Supplementary Table 18). pRH2502 (a vector expressing an inactivated version of the *cas9* DNA sequence) and pRH2521 (an sgRNA-expressing vector whose expression is regulated by the TetR-regulated smyc promoter $P_{myc1tetO}$) were introduced into the bacteria to obtain colonies of the knockdown strains. For selection, 25 µg/ml kanamycin and 50 µg/ml hygromycin were used for ATCC13950, and 100 µg/ml kanamycin and 50 µg/ml hygromycin were used for the clinical strains. The growth assay of the knockdown strains was performed in 7H9/ADC broth containing the above-mentioned concentrations of kanamycin and hygromycin under aerobic conditions at 37 °C with humidification (APM-30D incubator, Astec, Tokyo, Japan). During the assay, the cultures were supplemented with anhydrotetracycline (aTc) to a final concentration of 50 ng/ml (as a concentration that is not toxic to ATCC13950 and M.i.27) or 200 ng/ml (for the other clinical strains). aTc was added repeatedly every 48 h to maintain the induction of dCas9 and sgRNAs in experiments that extended beyond 48 h.²⁹ An aliquot (4 µl) of the culture and its diluents were streaked onto 7H10/OADC agar plates containing kanamycin and hygromycin, and the CFUs were counted. Growth inhibition was evaluated as the ratio of the CFUs of knockdown strains compared with those of the vector control strains lacking the guide RNA sequence. To confirm the downregulation of gene expression in the knockdown strains, quantitative reverse transcription PCR (qRT-PCR) was performed. Bacteria were cultured in 10 ml of 7H9/ADC/Tween 80 containing kanamycin and hygromycin until mid-log phase (OD_{600} approximately 0.4–0.6). The cultures were supplemented with aTc every 48 h, as in the growth assay for the knockdown strains. One day after the second addition of aTc, RNA was extracted by bead-beating six times at 5000 rpm for 30 s with cooling (Micro Smash MS 100R Cell Disruptor; Tomy Seiko, Tokyo, Japan) and cDNA was synthesized with the Direct-zol™ RNA Miniprep (Zymo Research, Orange, CA, USA) and ReverTra Ace qPCR RT Master Mix with gDNA Remover (Toyobo, Osaka, Japan). qRT-PCR was performed with the CFX Connect Real-time PCR Detection System (Bio-Rad, Hercules, CA, USA), according to the manufacturer's instructions. The data were normalized to the expression of *sigA*.

Statistical analyses

The cut-off of fold-change ($\log_2FC$) to compare genetic requirements by TnSeq resampling analysis was set as adjusted *P*-values less than 0.05. Pearson's correlation test was performed between the fitness costs ($\log_2FC$ calculated by the resampling assay), efficiency of knockdown (assessed by qRT-PCR) and growth suppression in the knockdown strains. Statistical analyses of the growth curves were performed using a 4-parameter logistic model (logistic 4P model), Wilcoxon two-sample test, and a one-way analysis of variance (ANOVA) followed by Steel's multiple comparisons test using the JMP software program (SAS Institute Inc., Cary, NC, USA). Data are presented as the mean ± SD. Statistical significance was set as *P* values less than 0.05.

Availability of data and materials

The datasets used or analyzed in this study are available from the corresponding author upon reasonable request.

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Additional information

Ethics approval and consent to participate

All experimental protocols were approved by the Institutional Review Board of Niigata University (approval numbers: 2019-0020 for the use of human tissue samples and SA00506 for experiments on live vertebrates). All experiments on humans and human samples were conducted in accordance with relevant guidelines and regulations. Informed consent was obtained from all patients. All experiments on animals were conducted in accordance with the relevant guidelines and regulations, and the study complied with the ARRIVE guidelines.

Consent for publication

According to the protocols approved by the Institutional Review Board of Niigata University mentioned above, consent for publication was obtained from the patients enrolled in this study.

Author contributions

Y.T. conceived, designed and performed the experiments. Y.O., A.N., and S.M. contributed reagents or materials. Y. Morishige., Y. Minato, and A.D.B. contributed analytical tools. Y. Morishige., Y. Minato, A.D.B., and S.M. provided critical suggestions throughout the preparation of the manuscript. Y.T. wrote the manuscript. All authors read and approved the final manuscript.

Additional files

Supplementary Information [↗](#)

Supplementary Table 1 [↗](#)

Supplementary Table 2 [↗](#)

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Supplementary Table 11 [↗](#)

Supplementary Table 12 [↗](#)

Supplementary Table 13 [↗](#)

Supplementary Table 14 [↗](#)

Supplementary Table 15 [↗](#)

Supplementary Table 16 [↗](#)

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Supplementary Table 18 [↗](#)

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Reviewer #1 (Public review):

Summary:

In this descriptive study, Tateishi et al. report a Tn-seq based analysis of genetic requirements for growth and fitness in 8 clinical strains of *Mycobacterium intracellulare* (Mi), and compare the findings with a type strain ATCC13950. The study finds a core set of 131 genes that are essential in all nine strains, and therefore are reasonably argued as potential drug targets. Multiple other genes required for fitness in clinical isolates have been found to be important for hypoxic growth in the type strain.

Strengths:

The study has generated a large volume of Tn-seq datasets of multiple clinical strains of Mi from multiple growth conditions, including from mouse lungs. The dataset can serve as an important resource for future studies on Mi, which despite being clinically significant remains a relatively understudied species of mycobacteria.

Weaknesses:

The primary claim of the study that the clinical strains are better adapted for hypoxic growth is yet to be comprehensively investigated. However, this reviewer thinks such an investigation would require a complex experimental design and perhaps forms an independent study.

<https://doi.org/10.7554/eLife.99426.2.sa3>

Reviewer #4 (Public review):

Summary:

In this study Tateishi et al. used TnSeq to identify 131 shared essential or growth defect-associated genes in eight clinical MAC-PD isolates and the type strain ATCC13950 of *Mycobacterium intracellulare* which are proposed as potential drug targets. Genes involved in gluconeogenesis and the type VII secretion system which are required for hypoxic pellicle-type biofilm formation in ATCC13950 also showed increased requirement in clinical strains

under standard growth conditions. These findings were further confirmed in a mouse lung infection model.

Strengths:

This study has conducted TnSeq experiments in reference and 8 different clinical isolates of *M. intracellulare* thus producing large number of datasets which itself is a rare accomplishment and will greatly benefit the research community.

Weaknesses:

- (1) A comparative growth study of pure and mixed cultures of clinical and reference strains under hypoxia will be helpful in supporting the claim that clinical strains adapt better to such conditions. This should be mentioned as future directions in the discussion section along with testing the phenotype of individual knockout strains.
- (2) Authors should provide the quantitative value of read counts for classifying a gene as "essential" or "non-essential" or "growth-defect" or "growth-advantage". Merely mentioning "no insertions in all or most of their TA sites" or "unusually low read counts" or "unusually high low read counts" is not clear.
- (3) One of the major limitations of this study is the lack of validation of TnSeq results with individual gene knockouts. Authors should mention this in the discussion section.

<https://doi.org/10.7554/eLife.99426.2.sa2>

Reviewer #5 (Public review):

Summary:

In the research article, "Functional genomics reveals strain-specific genetic requirements conferring hypoxic growth in *Mycobacterium intracellulare*" Tateshi et al focussed their research on pulmonary disease caused by *Mycobacterium avium-intracellulare* complex which has recently become a major health concern. The authors were interested in identifying the genetic requirements necessary for growth/survival within host and used hypoxia and biofilm conditions that partly replicate some of the stress conditions experienced by bacteria in vivo. An important finding of this analysis was the observation that genes involved in gluconeogenesis, type VII secretion system and cysteine desulphurase were crucial for the clinical isolates during standard culture while the same were necessary during hypoxia in the ATCC type strain.

Strength of the study:

Transposon mutagenesis has been a powerful genetic tool to identify essential genes/pathways necessary for bacteria under various in vitro stress conditions and for in vivo survival. The authors extended the TnSeq methodology not only to the ATCC strain but also to the recently clinical isolates to identify the differences between the two categories of bacterial strains. Using this approach they dissected the similarities and differences in the genetic requirement for bacterial survival between ATCC type strains and clinical isolates. They observed that the clinical strains performed much better in terms of growth during hypoxia than the type strain. These in vitro findings were further extended to mouse infection models and similar outcomes were observed in vivo further emphasising the relevance of hypoxic adaptation crucial for the clinical strains which could be explored as potential drug targets.

Weakness:

The authors have performed extensive TnSeq analysis but fail to present the data coherently. The data could have been well presented both in Figures and text. In my view this is one of the major weakness of the study.

<https://doi.org/10.7554/eLife.99426.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Tateishi et al. report a Tn-seq-based analysis of genetic requirements for growth and fitness in 8 clinical strains of Mycobacterium intracellulare (Mi), and compare the findings with a type strain ATCC13950. The study finds a core set of 131 genes that are essential in all nine strains, and therefore are reasonably argued as potential drug targets. Multiple other genes required for fitness in clinical isolates have been found to be important for hypoxic growth in the type strain.

Strengths:

The study has generated a large volume of Tn-seq datasets of multiple clinical strains of Mi from multiple growth conditions, including from mouse lungs. The dataset can serve as an important resource for future studies on Mi, which despite being clinically significant remains a relatively understudied species of mycobacteria.

Thank you for reviewing our manuscript and finding the significance of our data.

Weaknesses:

The paper lacks clarity in data presentation and organization. For example, some of the key data on cfu counts of clinical Mi strains in a mouse model can be presented along with the Tn-seq dataset in Figure 6, the visualization of which can be improved with volcano plots. etc. Improvement in data visualization is perhaps necessary throughout the paper.

Thank you for the comment on the data presentation of *in vivo* studies. We previously revealed the time-course data on CFUs, animal survival, and tissue pathology from the pure strains (Tateishi Y. BMC Microbiol. 2023; new Ref #22) . Based on these data, we assumed that we would be able to harvest sufficient number of colonies from mice infected with M.i.27 or M.i.198, and we performed *in vivo* TnSeq studies using these two strains. We have referred to our previous publication (new Ref #22) on the virulence of MAC-PD strains used in this study for mice in the revised manuscript (page12, line 212).

The data of CFU counts were shown in new Supplementary Fig. 3b. In the manuscript text, we explained as follows (page 12, lines 212-216): “The time course of the changes in the bacterial burden showed a pattern similar to those of the wild-type strains M.i.198 and M.i.27, respectively, except that it was not possible to harvest sufficient colonies (as few as 104/mouse) in the few mice infected with the M.i.27 Tn mutant strain in week 8 and week 16 (page 12, lines 212-216; new Supplementary Fig. 3b, new Supplementary Table 8)”.

Regarding the suggestion to include volcano plots, we appreciate the proposal but chose not to adopt this format, as the main aim of this study was to identify genes commonly required for *in vitro* and *in vivo* fitness across multiple *M. intracellulare* strains, rather than to highlight differential genetic requirements within a single strain. Volcano plots are useful for visualizing differential values and significance for a single dataset but are less suited for cross-strain comparisons of shared gene sets. Our approach is aligned with the methodology used by Cary et al. (PLoS Pathog. 2018; new Ref#8), who similarly focused on identifying conserved genetic requirements across *M. tuberculosis* genotypes without employing volcano plots.

[References]

Tateishi, Y. et al. Virulence of *Mycobacterium intracellulare* clinical strains in a mouse model of lung infection - role of neutrophilic inflammation in disease severity. BMC Microbiol 23, 94 (2023).

Carey, A.F. et al. TnSeq of *Mycobacterium tuberculosis* clinical isolates reveals strain-specific antibiotic liabilities. PLoS Pathog 14, e1006939 (2018).

The primary claim of the study that the clinical strains are better adapted for hypoxic growth is not well-supported by the data presented in Figure 7.

Thank you for the comments on the difference of adaptation for hypoxic growth between ATCC13950 and clinical MAC-PD strains. To clarify, growth rates shown in Figure 7 were calculated at the inflection point (midpoint) of the growth curves, which were modeled using a four-parameter logistic (4P logistic) model. As described in the Discussion, we found the pattern of hypoxic adaptation characteristics of the clinical MAC-PD strains from the growth curve forms. Taking into consideration the impact of growing bacteria on the disease progression of MAC-PD, the slow growth with early entry to log phase implicates the continuous impact on the infected hosts during logarithmic bacterial growth, which may be involved in the persistent and steadily progressive illness of MAC-PD for years in humans.

Unlike time-lapse imaging assay, the completely seamless sampling of culture for CFU assay is impossible. Nevertheless, we collected sufficient timepoints to allow reliable curve fitting with the 4P logistic model and thus consider our growth data to represent a valid approximation of continuous growth dynamics.

Regarding the suggestion of mixed culture experiments, we agree that such studies could be informative. However, co-culture conditions introduce additional variables, including inter-strain competition or synergy, which can obscure the specific contributions of hypoxic adaptation in each strain. Therefore, we believe that the current approach using monoculture growth curves under defined oxygen conditions offers a clearer interpretation of strain-specific hypoxic responses.

The title of the paper is misleading as the study doesn't provide any mechanistic aspect of hypoxic adaptation in Mi.

Thank you for the comment on the article title. We admit that this paper does not directly reveal the mechanism of hypoxic adaptation in *M. intracellulare* strains but provides the data on the different pattern of hypoxic adaptation between *M. intracellulare* strains in relation to the difference of genetic requirements. Therefore, we revised the title as "Functional genomics reveals strain-specific genetic requirements conferring hypoxic growth in *Mycobacterium intracellulare*"

Reviewer #2 (Public Review):

Summary:

*In the study titled "Functional genomics reveals the mechanism of hypoxic adaptation in nontuberculous mycobacteria" by Tateishi et al., the authors have used TnSeq to identify the common essential and growth-defect-associated genes that represent the genomic diversity of clinical *M. intracellulare* strains in comparison to the reference type strain. By estimating the frequency of Tn insertion, the authors speculate that genes involved in gluconeogenesis, the type VII secretion system, and cysteine desulfurase are relatively critical in the clinical MAC-PD strains than in the type strain, both for the extracellular survival and in a mouse lung infection model.*

*Based on their analysis, the authors proposed to identify the mechanism of hypoxic adaptation in nontuberculous mycobacteria (NTM) which offer promising drug targets in the strains causing clinical *Mycobacterium avium-intracellulare* complex pulmonary disease (MAC-PD).*

Strengths:

*A major strength of the manuscript is the performance of the exhaustive set of TnSeq experiments with multiple strains of *M. intracellulare* during in vitro growth and animal infection.*

Thank you for reviewing our manuscript and acknowledging the performance of producing datasets in this study.

Weaknesses:

(1) The study suffers from the authors' preconceived bias toward a small subset of genes involved in hypoxic pellicle formation in ATCC13950.

Thank you for the comment regarding a potential bias toward a small subset of genes involved in hypoxic pellicle formation in ATCC13950. The rationale for the importance of hypoxic pellicle genes in clinical MAC-PD strains is that the profiles of genetic requirements in each bacterial strain reflect the adaptation to the environment in which each strain lives. When the strains are placed in a special environment, they can adapt to the situation by altering the profiles of genetic requirements, resulting in the remodeling of metabolic pathways.

In this study, we found that several of these pellicle-associated genes also showed increased genetic requirement in the clinical MAC-PD strains, suggesting a possible overlap in hypoxic adaptation mechanisms. We did not insist that clinical MAC-PD strains showed an increase of genetic requirements in all genes of hypoxic pellicle formation. Except for the gene sets involved in hypoxic pellicle formation in ATCC13950, almost no global information has been revealed on the pathogenesis of nontuberculous mycobacterial disease, which differs from the case of tuberculosis. Along with this finding, we investigated the effect of gene silencing on bacterial growth and preferential hypoxic adaptation observed by growth kinetics in clinical MAC-PD strains compared to ATCC13950. At first glance, to focus on the gene sets of hypoxic pellicle formation seems to be "biased", but we proceeded this research step by step based on our achievements. We consider these data provide valuable information on the pathogenesis of MAC-PD by clinical MAC-PD strains.

We have added the description of the rationale for the importance of hypoxic pellicle genes in clinical MAC-PD strains in the revised manuscript (page 9, lines 148-155).

(2) An important set of data with the ATCC13950 reference strain is missing in the mouse infection study. In the absence of this, it is difficult to establish whether the identified genes are critical for infection/intracellular proliferation, specifically in the clinical isolates that are relatively more adapted for hypoxia.

Thank you for the comment on the necessity of setting ATCC13950 as a control strain of mouse TnSeq experiment. To set ATCC13950 as a control strain in mouse infection experiments would be ideal. However, we proved that ATCC13950 is eliminated within 4 weeks of infection (Tateishi Y. BMC Microbiol. 2023; new Ref#22). That means, it is impossible to perform *in vivo* TnSeq study due to the inability to harvest sufficient number of colonies.

[Reference]

Tateishi, Y. et al. Virulence of *Mycobacterium intracellulare* clinical strains in a mouse model of lung infection - role of neutrophilic inflammation in disease severity. BMC Microbiol 23, 94 (2023).

(3) Statistical enrichment analysis of gene sets by GSEA wrongly involves genes required for hypoxic pellicle formation in ATCC13950 together with the gene sets found essential in the clinical MAC-PD strains, to claim that a significant % of genes belong to hypoxia-adaptation pathways. It could be factually incorrect because a majority of these might overlap with those found critical for the *in vitro* survival of MAC-PD strains (and may not be related to hypoxia).

Thank you for the suggestion on the re-analysis of gene enrichment analysis of genes required for M.i.27 and M.i.198 *in vivo* infection, individually with genes involved in hypoxic pellicle formation in ATCC13950 and with those showing increased genetic requirements in clinical MAC-PD strains compared to ATCC13950.

About 50% (92 and 94 out of 181 genes through Day 1 to Week 16 and Week 4 to Week 16 of infection) and 40% (70 and 79 genes out of 179 through Day 1 to Week 16 and Week 4 to Week 16 of infection) of genes required for hypoxic pellicle formation in ATCC13950 were listed as enriched in genes required for mouse lung infection in M.i.27 and M.i.198, respectively. In addition, about 42% (54 and 56 out of 128 genes through Day 1 to Week 16 and thorough Week 4 to Week 16 of infection) and 40% (79 and 68 out of 179 genes through Day 1 to Week 16 and through Week 4 to Week 16 of infection) of genes showing increased requirements in clinical MAC-PD strains compared to ATCC13950 were listed as enriched in genes required for mouse lung infection in M.i.27 and M.i.198, respectively.

These data indicate that about 40-50% genes required for *in vitro* hypoxic pellicle formation are shared with the genes required for *in vivo* bacterial growth, and that about 40% strain-dependent/accessory essential genes are shared with the genes required for *in vivo* bacterial growth. Thus, the genes required for the growth of M.i.27 and M.i.198 in mouse lungs are enriched individually with those involved in hypoxic pellicle formation in ATCC13950, and with the gene sets found critical for *in vitro* growth. We have added the description of the reanalyzed data of GSEA in the manuscript (pages 16-17, lines 287-290). And the details of reanalyzed data of GSEA have been shown in Supplementary Fig. 5 and 6 as well as Supplementary Tables 15 and 16.

(4) Validation of mouse infection experiments with individual mutants is missing.

Thank you for the suggestion on the validation of the TnSeq hit genes on the *in vivo* survival. We acknowledge the importance of validating the TnSeq-hit genes by constructing knockout mutants. We have recently succeeded in constructing the vectors for making knockout strains

of *M. intracellulare* (Tateishi. Microbiol Immunol. 2024). We will proceed to the infection experiment of knockout mutants by using our system for constructing them.

[Reference]

Tateishi Y. et al. Construction of knockout mutants in *Mycobacterium intracellulare* ATCC13950 strain using a thermosensitive plasmid containing negative selection marker *rpsL*. Microbiol Immunol 68, 339-347 (2024).

(5) Phenotypes with TnSeq and CRISPRi-based KD exhibit poor correlation with misleading justifications by the authors.

Thank you for the comment on the issue of inconsistent results between TnSeq and CRISPR-i based knockdown. We acknowledge that some inconsistencies were observed, particularly among strain-dependent/accessory essential or growth-defect-associated genes. By contrast, we found consistent data between TnSeq and CRISPR-i based knockdown results among universal essential genes such as *glcB*, *inhA*, *gyrB* and *embB*. Although the mechanism has not been fully proven yet, we consider that such inconsistent phenotypes with TnSeq and CRISPR-based knockdown may be related to the recently revealed bypass mechanism of gene essentiality which is characteristically observed in strain-specific/accessory essential genes (Rosconi F. Nat Microbiol. 2022; new Ref#14). They suggested this bypass mechanism of gene essentiality in strain-dependent/accessory essential or growth-defect-associated genes from the ‘forced-evolution experiments’ of 36 clinical *Streptococcus pneumoniae* strains. For example, knockout mutants are successfully recovered from transformation experiments targeting strain-specific/accessory essential genes in TnSeq such as cytidine monophosphate kinase *cmk*, formate tetrahydrofolate ligase *fhs* and farnesyl-diphosphate synthase *fpp*. The bypassing of gene essentiality can be suggested by observing suppressor mutations and synthetic lethality in knockout strains. By contrast, universal essential genes fulfill the following three categories: i) high levels of conservation within and often across species, ii) limited genetic diversity, and iii) high and stable expression levels. Consequently, the universal essential genes are rigid, largely immutable key components to an organism’s survival. In the universal essential genes, the knockout recovery fails as shown by no colonies or only appearance of merodiploids. Taking into consideration such bypass mechanism of gene essentiality in strain-dependent/accessory essential genes, the lower effect of gene silencing of strain-dependent/accessory essential genes on bacterial growth may reflect pathway rewiring that helps the bacterial growth under suppression of the target gene expression.

We have added the description of the possible reason for inconsistency between TnSeq and CRISPR-i results in the Result and Discussion in the revised manuscript (page 21, lines 367-376; pages 28-29, lines 489-519).

[Reference]

Rosconi, F. et al. A bacterial pan-genome makes gene essentiality strain-dependent and evolvable. Nat Microbiol 7, 1580–1592 (2022).

In summary, this study is unable to provide mechanistic insights into why and how different MAC-PD mutant strains exhibit differential survival (in vitro and in animals) and adaptation to hypoxia. It remains to understand why the clinical strains show better adaptation to hypoxia and what is the impact of other stresses on their growth rates.

Thank you for the comments on the issue of being unable to prove the mechanism of MAC-PD pathogenesis and adaptation to hypoxia. We admit that the original manuscript did not provide the apparent reason and mechanism of MAC-PD pathogenesis and adaptation to hypoxia. Following the comment, we have modified the manuscript title as “Functional

genomics reveals strain-specific genetic requirements conferring hypoxic growth in *Mycobacterium intracellulare*".

However, we revealed the diversity of genetic requirements among the genus *M. intracellulare* including the type strain ATCC13950 and clinical MAC-PD strains. We revealed the characteristics of genetic requirements in clinical MAC-PD strains as increased genetic requirements of gluconeogenesis, type VII secretion system and cysteine desulfurase, the former two of which are also required in hypoxic pellicle formation in ATCC13950. Along with this, we demonstrated the difference of growth behavior under hypoxia between clinical MAC-PD strains and ATCC13950. Overall, we consider that we could provide the basic information suggesting the involvement of difference of genetic requirements among strains in the pathogenesis of MAC-PD.

Reviewer #3 (Public Review):

Summary:

The study by Tateishi et al. utilized TnSeq in nine genetically diverse M. intracellulare strains, identifying 131 common essential and growth-defect-associated genes across those strains, which could serve as potential drug targets. The authors also provided an overview of the differences in gene essentiality required for hypoxic growth between the reference strain and the clinical strains. Furthermore, they validated the universal and accessory/strain-dependent essential genes by knocking down their expression using CRISPRi technique. Overall, this study offers a comprehensive assessment of gene requirements in different clinical strains of M. intracellulare.

Thank you for reviewing our manuscript and finding the significance of our data.

(1) The rationale for using ATCC13950 versus clinical strains needs to be clarified. The reference strain ATCC13950 was obtained from the abdominal lymph node of a patient around 10 years ago and is therefore considered a clinical strain that has undergone passages in vitro. How many mutations have accumulated during these in vitro passages? Are these mutations significant enough to cause the behavior of ATCC13950 to differ from other recently sampled clinical strains? From the phylogenetic tree, ATCC13950 is located between M018 and M.i.27. Did the authors observe a similarity in gene essentiality between ATCC13950 and its neighbor strains? What is the key feature that separates ATCC13950 from these clinical strains? The authors should provide a strong rationale for how to interpret the results of this comparison in a clinical or biological context.

Thank you for the comments on the rationale for using ATCC13950 versus clinical strains and the key feature that separates ATCC13950 from clinical MAC-PD strains.

ATCC13950 was isolated in 1949, (not around 10 years ago) from 34-month-old female abdominal lymph node (Cuttino. Am J Pathol 1949; new Ref#11). Of note, the clinical background of the patient infected with ATCC13950 is quite different from the patients with MAC-pulmonary disease (MAC-PD), the incidence rate of which has been increasing worldwide without predisposing immunological disorders. ATCC13950 has been regarded as a type strain of genus *M. intracellulare* historically. And ATCC13950 is the first *M. intracellulare* strain to be sequenced in 2012 (Kim. J Bacteriol 2012; new Ref#59).

The rationale for using ATCC13950 versus clinical MAC-PD strains is to find the answer to the question whether the essential genes and genetic requirements are similar or different between clinical MAC-PD strains and historical type strain ATCC13950. So far, there are two reports on TnSeq that compare genetic requirements between clinical mycobacterial strains and the type strains, one of which is *M. tuberculosis* (Carey AF. PLoS Pathogens. 2018; new

Ref#8) and the other is *M. abscessus* (Akusobi C. mBio. 2025; new Ref#9, published after submission of our manuscript). They reported the difference and diversity of genetic requirements between clinical strain and type strains such as *M. tuberculosis* H37Rv and *M. abscessus* ATCC19977. We have added the mention of these previous reports to explain the rationale for setting the type strain ATCC13950 as a referential control strain. (page 5, lines 83-87)

The genetic and functional analysis of clinical MAC-PD strains has not been conducted for a long time. In 2021, we have revealed the genomic diversity between clinical MAC-PD and ATCC13950 by comparative genomic analysis (Tateishi BMC Microbiol. 2021; new Ref#5). Except for our TnSeq study of ATCC13950 (Tateishi Sci Rep 2020; new Ref#10), no functional analysis has been conducted in clinical *M. intracellulare* strains. On our research stream of clinical MAC-PD strains, we expected that we could reveal the functional genomic characteristics of clinical MAC-PD strains by setting ATCC13950 as a referential control strain for analyzing TnSeq data.

It seems an interesting viewpoint to consider the relationship between accumulation of mutations by *in vitro* passages during prolonged periods from first isolation in ATCC13950, and the difference of phenotypes between ATCC13950 and recently sampled clinical MAC-PD strains. However, there are no time-course samples of ATCC13950 isolates available. Therefore, we can neither investigate how many mutations have accumulated in a time-course manner, nor evaluate how much the accumulated mutations influence the phenotype in ATCC13950. It can be expected that the accumulation of *in vitro* mutations may cause the behavior of ATCC13950 different from clinical MAC-PD strains. However, it is to be elucidated yet which kinds of factors contribute to the characteristics of ATCC13950 that differ from clinical MAC-PD strains specifically.

It seems an interesting viewpoint to investigate the similarity of gene essentiality between genetical neighbor strains. However, we focused on the overview of the profiles of gene essentiality in clinical MAC-PD strains compared to ATCC13950. Thus, it was out of scope to elucidate the details of gene essentiality in each genetic phylogeny that clinical MAC-PD strains belong. The overview of phylogenetic trees should be referred to our previous publication on the comparative genomic analysis of 55 strains (Tateishi Y. BMC Microbiol. 2021; new Ref#5, new Supplementary Fig. 1), and we have shown Fig. 1 as the extracted phylogenetic tree of subject strains. To elucidate the details of gene essentiality in each genetic clade, it would be necessary to include a considerable number of strains that we used for comparative genomic analysis in 2021 (Tateishi Y. BMC Microbiol. 2021; new Ref#5). Furthermore, it would be necessary to set a referential control strain other than ATCC13950 for comparing gene essentiality. So far, it is not the highest priority for us to elucidate the similarity of gene essentiality between phylogenetic clades in detail, and such investigation will be planned as a future study.

The key features that separate ATCC13950 and clinical MAC-PD strains have not been proved yet, in contrast to the case of *M. tuberculosis* such as mutations in the gene of the response regulator PhoPR in the type strain H37Rv vs most clinical strains. However, the features that separate ATCC13950 and clinical MAC-PD strains may not be explained by a single genetic factor but may be explained by complicated factors such as epigenetic and/or regulatory factors. For example, the reason for the weakened virulence of H37Ra compared to H37Rv has not been able to be explained by simple genetic differences (Brosch R. Infect Immun. 1999).

In summary, we set the historical type strain ATCC13950 which is derived from infant abdominal lymphadenitis as a referential control strain for TnSeq analysis, because we intended to reveal the characteristics of clinical MAC-PD strains in terms of the gene essentiality and genetic requirements by comparing the clinical MAC-PD strains with the non-MAC-PD reference strain. We consider that the profiles of gene essentiality and genetic

requirements specific to clinical MAC-PD strains confer the pathogenesis in an increasing number of MAC-PD patients worldwide without predisposing immunological disorders.

[References]

Cuttino, J.T. & Mc, C.A. Pure granulomatous nocardiosis, a new fungus disease distinguished by intracellular parasitism; a description of a new disease in man due to a hitherto undescribed organism, *Nocardia intracellularis*, n. sp., including a study of the biologic and pathogenic properties of this species. *Am J Pathol* 25, 1-47 (1949).

Kim, B.J. et al. Complete genome sequence of *Mycobacterium intracellulare* clinical strain MOTT-64, belonging to the INT1 genotype. *J Bacteriol* 194, 3268 (2012).

Carey, A.F. et al. TnSeq of *Mycobacterium tuberculosis* clinical isolates reveals strain-specific antibiotic liabilities. *PLoS Pathog* 14, e1006939 (2018).

Akusobi, C. et al.. Transposon-sequencing across multiple *Mycobacterium abscessus* isolates reveals significant functional genomic diversity among strains. *mBio* 6, e0337624 (2025).

Tateishi, Y. et al. Comparative genomic analysis of *Mycobacterium intracellulare*: implications for clinical taxonomic classification in pulmonary *Mycobacterium avium-intracellulare* complex disease. *BMC Microbiol* 21, 103 (2021).

Tateishi, Y. et al. Genome-wide identification of essential genes in *Mycobacterium intracellulare* by transposon sequencing - Implication for metabolic remodeling. *Sci Rep* 10, 5449 (2020)

Brosch R. et al. Genomic analysis reveals variation between *Mycobacterium tuberculosis* H37Rv and the attenuated *M. tuberculosis* H37Ra strain. *Infect Immun.* 67, 5768-74 (1999).

(2) Regarding the 'nine representative strains of *M. intracellulare* with diverse genotypes in this study,' how were these nine strains selected? To what extent do they represent the genetic diversity of the *M. intracellulare* population? A phylogenetic tree illustrating the global genetic diversity of the *M. intracellulare* population, with these strains marked on it, would be important to demonstrate their genetic representativeness.

Thank you for the comments on the selection of 9 subject strains. We selected the 9 strains based on the phylogenetic tree we published in 2021 (*BMC Microbiol* 2021; new Ref#5). We have shown the global phylogenetic tree of the *M. intracellulare* population in new supplementary Fig. 1. We have selected 4 or 5 strains from the major two groups (typical *M. intracellulare* group and *M. paraintracellulare*-*M. indicus pranii* group) for this TnSeq study, respectively.

[Reference]

Tateishi, Y. et al. Comparative genomic analysis of *Mycobacterium intracellulare*: implications for clinical taxonomic classification in pulmonary *Mycobacterium avium-intracellulare* complex disease. *BMC Microbiol* 21, 103 (2021).

(3) The authors observed a considerable amount of differential gene requirements in clinical strains. However, the genetic underpinning underlying the differential requirement of genes in clinical strains was not investigated or discussed. Because *M. intracellulare* has a huge number of accessory genes, the authors should at least check whether the differential requirement could be explained by the existence of a second copy of functional analogous genes or duplications.

Thank you for the comments on the effect of gene duplication on the change of genetic requirements between strains. Following the comments, we conducted blast search for the 162 genes showing increased Tn insertion reads in each subject strain. We found that M019 has duplicate genes of OCU_RS44705 coding adenosylhomocysteinase (LOCUS_42940: ahcY_1, LOCUS_21000: ahcY_2). However, there were no duplicate genes found in the remaining 161 genes showing increased Tn insertion reads.

From these results, we consider that gene duplication has minor effects on the change of genetic requirements between strains. Rather, sequence differences and accessory genes may play a key role in determining the difference of genetic requirements.

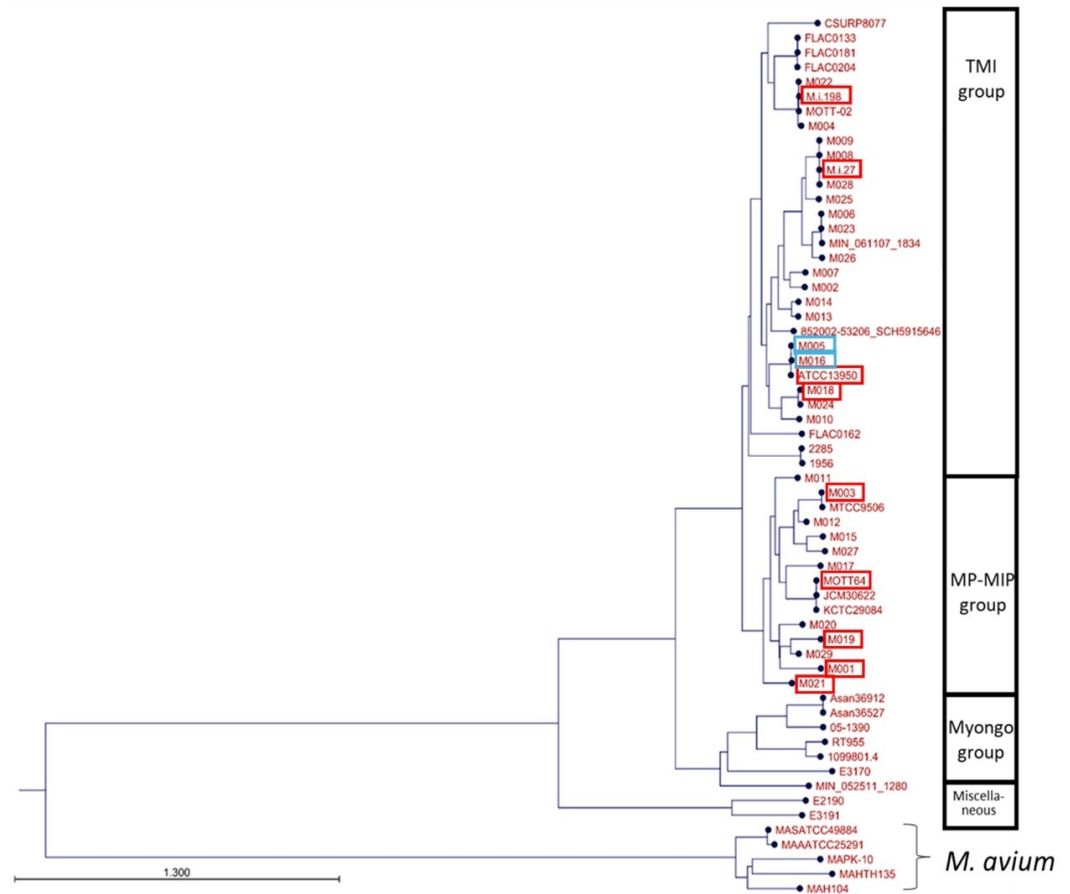
We have added a description of the above-mentioned result in the Result section (pages11-12, lines 191-199).

(4) Growth in aerobic and hypoxic conditions: The authors concluded that clinical strains are better adapted to hypoxia, as reflected by their earlier entry into the log phase. They presented the 'Time at midpoint' and 'Growth rate at midpoint.' However, after reviewing the growth curves, I noticed that ATCC13950 had a longer lag phase compared to other strains under hypoxic conditions, and its phylogenetic neighbor M018 also had a longer lag phase. Hence, I do not believe a conclusion can be drawn that clinical strains are better adapted to hypoxia, as this behavior could be specific to a particular clade. It's also possible that the ATCC13950 strain has adapted to aerobic growth. I would suggest that the authors include growth curves in the main figures. The difference in 'Time at midpoint' could be attributed to several factors, and visualizing the growth curves would provide additional context and clarity.

Thank you for the comments on the possibility of genotypes as a determinant of growth pattern in *M. intracellulare*. Following the comments, we performed aerobic and hypoxic growth assay in the two strains (M005 and M016) that neighbor ATCC13950.

Author response image 1.

The phylogenetic relationship between M005, M016 and ATCC13950. The former two strains are squared in blue.



M005 reached midpoint later than ATCC13950 both in aerobic and hypoxic conditions. By contrast, M016 reached midpoint three quarters earlier than ATCC13950 under hypoxic conditions. The growth rate was not significantly different between M005, M016 or ATCC13950 under either aerobic or hypoxic conditions, although *P*-value of M005 vs ATCC13950 was 0.0512 under aerobic conditions on Steel's multiple comparisons test.

From the data of growth pattern in M005 and M016, we suggest that in addition to gene essentiality, genotypes may have some impact on the bacterial growth pattern under hypoxia; however, since there was a significant difference in the timing of hypoxic adaptation among ATCC13950 and its neighbor strains, bacterial growth pattern under hypoxia is considered to be determined by multiple factors such as genetic requirements and unproven regulatory systems. Taking into consideration that there are lots of genetically diverse strains other than ATCC13950 clade, many clinical MAC-PD strains are possibly better adapted to hypoxia.

Responding to the reviewer's recommendation, we have added the description of the above-mentioned result in the revised manuscript (page 18, lines 313-322). And we have shown the data of growth curves of the original 9 subject strains in the new Fig 7. And we have added the data of the growth curves of M005 and M016 in new Supplementary Fig 7. Additionally, we have corrected the label of y-axis in new Fig. 7a and new Supplementary Fig. 7a and added the description as "Data are represented as CFUs in 4 μ l sample at each timepoint." in the Figure legends. (page 58, lines 1027-1028 and Supplementary Fig. 7a legend)

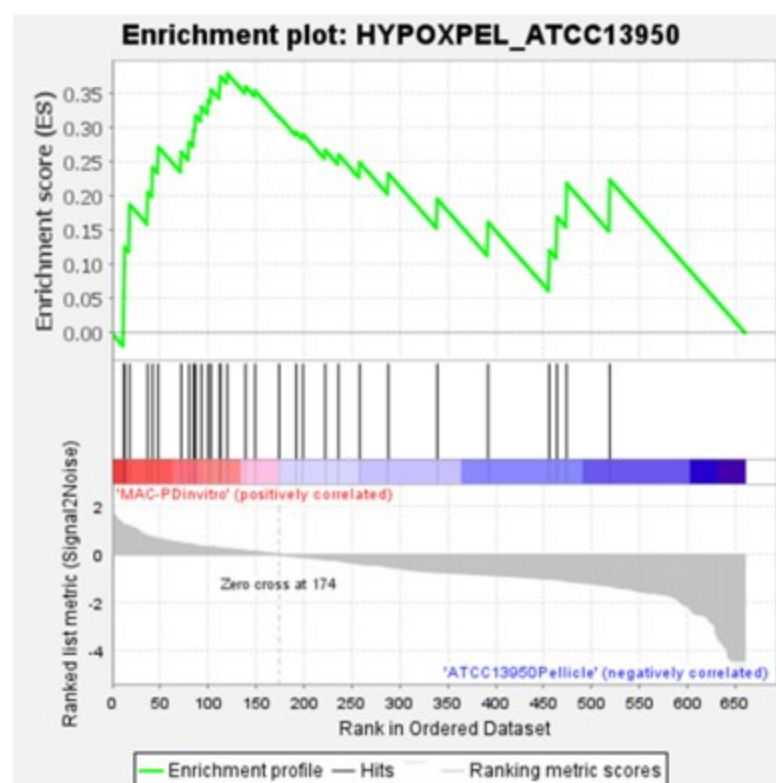
(5) *Lack of statistical statement: The authors emphasized the role of pellicle-formation-associated genes in strain-dependent essential and accessory essential genes. Additionally, the authors observed that 10% of the genes required for mouse infection are also required for hypoxic pellicle formation. However, these are merely descriptive*

statements. There is no enrichment analysis to justify whether pellicle-formation-associated genes are significantly enriched in these groups.

Thank you for the comments on the enrichment of pellicle-formation associated genes in strain-dependent essential and accessory essential genes. We performed GSEA and found that 9.1% (16 out of 175) genes were hit as core enrichment. Of them, 4 genes were hit commonly as genes showing increased genetic requirements analyzed by resampling plus HMM analyses including genes of phosphoenolpyruvate carboxykinase *pckA* (OCU_RS48660), type VII secretion-associated serine protease *mycP5* (OCU_RS38275), type VII secretion protein *eccC5* (OCU_RS38345) and glycine cleavage system amino-methyltransferase *gcvT* (OCU_RS35955).

Author response image 2.

Enrichment Score (ES)	0.37941608
Normalized Enrichment Score (NES)	1.9215446
Normalized p-value	0.0
FDR q-value	0.0
FWER p-value	0.0



We have added the description of GSEA result in the revised manuscript (page 8, lines 137-144; Supplementary Fig. 2; Supplementary Table 5).

Reviewer #1 (Recommendations For The Authors):

Tn-seq and hypoxia adaption in clinical isolates of M. intracellulare (Mi): The authors claim that clinical strains are better adapted to hypoxia because their genetic requirements for optimum fitness overlap with genetic requirements for fitness of the type strain under hypoxia. This is a reasonable hypothesis, but it has not been well-supported by the data presented in Figure 7. The growth rates (Figure 7b) of most of the clinical strains under hypoxia appear to be less than the type strain, although they all seem to grow better than the type strain under normoxia. Perhaps a continuous growth curve of each strain, both as pure and mixed cultures under these conditions will provide a clearer picture.

Thank you for the comments on the difference of adaptation of hypoxic growth between ATCC13950 and MAC-PD strains. To clarify, growth rates shown in Figure 7 were calculated at the inflection point (midpoint) of the growth curves, which were modeled using a four-parameter logistic (4P logistic) model. As described in the Discussion, we found the pattern of hypoxic adaptation characteristics of the clinical MAC-PD strains from the growth curve forms. Taking into consideration the impact of growing bacteria on the disease progression of MAC-PD, the slow growth with early entry to log phase implicates the continuous impact on the infected hosts during logarithmic bacterial growth, which may be involved in the persistent and steadily progressive illness of MAC-PD for years in humans.

Unlike time-lapse imaging assay, the completely seamless sampling of cultures for CFU assay is impossible. Nevertheless, we collected sufficient timepoints to allow reliable curve fitting with the 4P logistic model, and thus consider our growth data to represent a valid approximation of continuous growth dynamics.

Regarding the suggestion of mixed culture experiments, we agree that such studies could be informative. However, co-culture conditions introduce additional variables, including inter-strain competition or synergy, which can obscure the specific contributions of hypoxic adaptation in each strain. Therefore, we believe that the current approach using monoculture growth curves under defined oxygen conditions offers a clearer interpretation of strain-specific hypoxic responses.

In vivo studies: It is unclear how virulent the two clinical strains, Mi27 and Mi198 are in the mouse model. The CFU data in Figure S1b reports the bacterial burden of the Tn libraries of the two strains, of which the overall population of Mi27 library seems to be declining. Without any information on the CFU, animal survival, and tissue pathology from the pure strains, data from the library will have limited implications.

Thank you for the comments on the data presentation of *in vivo* studies. We previously revealed the time-course data on CFUs, animal survival, and tissue pathology from the pure strains (Tateishi Y. BMC Microbiol. 2023; new Ref#22). Based on these data, we assumed that we would be able to harvest sufficient number of colonies from mice infected with M.i.27 or M.i.198, and we performed *in vivo* TnSeq studies using these two strains. We have referred to our previous publication on the virulence of MAC-PD pure strains used in this study for mice in the revised manuscript (page 12, line 212; new Ref #22).

The data of CFU counts were shown in new Supplementary Figure 3b. In the manuscript text, we explained as follows (page 12, lines 212-216): “The time course of the changes in the bacterial burden showed a pattern similar to those of the wild-type strains M.i.198 and M.i.27, respectively (Tateishi Y. BMC Microbiol. 2023; new Ref#22), except that it was not possible to harvest sufficient colonies (as few as 104/mouse) in the few mice infected with the M.i.27 Tn mutant strain in week 8 and week 16 (new Supplementary Fig, 3b, new Supplementary Table 8)”. The decline of overall population of M.i.27 Tn mutant library strains in the infected lungs

can be explained by the lower virulence of M.i.27 pure strain that shows intermediate virulence phenotype than M.i.198 that shows high virulence phenotype.

[References]

Tateishi, Y. et al. Virulence of *Mycobacterium intracellulare* clinical strains in a mouse model of lung infection - role of neutrophilic inflammation in disease severity. BMC Microbiol 23, 94 (2023).

Data presentation: The manuscript suffers from a lack of clarity in data visualization and presentation, especially the Tn-Seq datasets. Panels describe the experimental workflow with a densely-worded paragraph, making it difficult to navigate through the major findings.

Thank you for the comments on the issue of Fig. 1b. Following the suggestion, we have modified the new Fig. 1b entitled “Strategy of the procedure of TnSeq analyses”.

Language: The paper should be extensively revised for language. Often the authors have mixed-up the terms like 'core' and 'accessory' 'genes' in lines 116-119 with 'core and accessory genomes' in Figure 2c, which is not even mentioned in the paper. It is further unclear how they identified 3153 and 5824 core and accessory genes, respectively, from 55 different strains of Mi. Line 251: ... "involved by confer..." needs revision. The terms "increased gene essentiality" and 'growth-defected associated genes" are very confusing. The essentiality of a gene is either absolute or conditional but is not quantitative. Similarly, 'growth-defect associated' can be replaced with a better phrase that alludes to fitness loss in the clone. Additional typos were found throughout the paper that need to be fixed.

Thank you for the comments on the issue of scientific words including “core and accessory genomes” and “gene essentiality” used in this study.

Based on Rosconi’s paper (Panel C of Fig. 1 in Nat Microbiol. 2022; new Ref#14), we used the phrases “accessory genome and core genome” as a meaning of a whole set of genes belonging to accessory and core genes. To avoid the confusion and keep consistency, we replaced the term “genomes” to “genes” in the revised manuscript.

In our previous comparative genomic analysis, we identified 3153 and 5824 core and accessory genes, respectively, from 55 different strains of *M. intracellulare* (Tateishi Y. BMC Microbiol. 2021; new Ref #5). To perform pangenomic analysis, we used the software Bacterial Pan-Genome Analysis tool (BGAP) (Narendrakumar NM. Sci Rep 2016).

We admit that gene essentiality is a qualitative but not a quantitative trait. We have corrected the term "increased gene essentiality" as "increased genetic requirements" throughout the manuscript.

We have used the term “growth-defect (GD)” based on the classification of gene essentiality calculated by the Hidden Markov Model (HMM) compiled by TRANSIT software (DeJesus. PLoS Comput Biol. 2015; new Ref#12). The HMM classifies genes as essential (ES), GD, non-essential (NE), growth-advantage (GA). GD means difficulties of growth (growth deficiency) in aerobic conditions *in vitro*, because Tn insertions are less frequent. The suggested phrases “fitness loss” or “less fit” may include the meaning of the comparison of two different conditions such as culture conditions exposed to a single bacterial strain. Since the HMM analysis is performed in data of a single strain in a specific bacterial condition, we consider that the phrase including “fitness” is somewhat unsuitable for describing the classification of gene essentiality. Thus, it is difficult for us to rephrase GD to the word that implies fitness levels between two conditions in a single bacterial strain.

[References]

Rosconi, F. et al. A bacterial pan-genome makes gene essentiality strain-dependent and evolvable. *Nat Microbiol* 7, 1580–1592 (2022).

Tateishi, Y. et al. Comparative genomic analysis of *Mycobacterium intracellulare*: implications for clinical taxonomic classification in pulmonary *Mycobacterium avium-intracellulare* complex disease. *BMC Microbiol* 21, 103 (2021).

Narendrakumar NM et al. BPGA- an ultra-fast pan-genome analysis pipeline. *Sci Rep* 2016 6, 24373 (2016).

DeJesus, M.A. et al. TRANSIT--A Software Tool for Himar1 TnSeq Analysis. *PLoS Comput Biol* 11, e1004401 (2015).

Reviewer #2 (Recommendations For The Authors):

Major Comments:

(1) Result 1 (Page 6-7): Common essential and growth-defect-associated genes representing the genomic diversity of M. intracellulare strains.

(1a) From Table S1, it is observed that the numbers of Tn-inserted TA sites significantly vary ($p > 0.05$) among biological replicates for each strain when compared with the reference strain ATCC13950. the authors should provide an explanation of how they overcame this variation in their analysis.

Thank you for the comment on the issue of a variable number of Tn-inserted TA sites among biological replicates for each strain of MAC-PD.

On TRANSIT software, we set the replicate option as Sum to combine read counts. And we used Beta-Geometric correction (BGC) to normalize the datasets to fit an “ideal” geometric distribution with a variable probability parameter ρ .

Following the comment, we have added the description on which option we used for handling the replicate data and normalization (page 36, lines 640-643).

(1b) Importantly, saturation in most of the strains is only ~50-60%. In such a case, there will be a high probability that Tn will not hit a nonessential region due to chance instead of selection (See DeJesus et al., mBio, 2017). It has been observed that the sequence pattern (GC)GNTANC(GC) is strongly associated with non-permissiveness. As shown earlier, the authors need to carefully look for the potential non-permissive sites before concluding the fate of a gene. Also, they should acknowledge the potential limitations of their approach due to the suboptimal level of saturation.

Thank you for the comment on the saturation of Tn mutant libraries. Our method of comparison of genetic requirements between strains are the same as a previous report that used duplicate Tn mutant libraries of clinical Mtb strains of different genotypes and triplicate Tn mutant libraries of H37Rv for identifying increased genetic requirements of clinical Mtb strains (Carey. *PLoS Pathog* 2018; new Ref#8). Our method is also based on the coauthor’s TnSeq study on H37Rv (Minato Y. *mSystems* 2019; new Ref#61). Moreover, by combining replicates, the saturation of our Tn mutant libraries became 62-79% as follows: ATCC13950: 67.6%, M001: 72.9%, M003: 63.0%, M018: 62.4%, M019: 74.5%, M.i.27: 76.6%, M.i.198: 68.0%, MOTT64: 77.6%, M021: 79.9%. That is, we calculated gene essentiality from the Tn mutant libraries with 62-79% saturation in each strain. The levels of saturation of transposon libraries in our study is similar to the very recent TnSeq analysis by Akusobi where 52-80%

saturation libraries (so-called “high-density” transposon libraries) are used for HMM and resampling analyses (Supplemental Methods Table 1[merged saturation] in Akusobi. mBio. 2025; new Ref#9). The saturation of Tn insertion in individual replicates of our libraries is also comparable to that reported by DeJesus (Table S1 in mBio 2017; new Ref#57). Thus, we consider that our TnSeq method of identifying essential genes and detecting the difference of genetic requirements between clinical MAC-PD strains and ATCC13950 is acceptable.

As the Reviewer indicates, there is non-permissive sequence pattern in mariner transposon mutagenesis. Using more than 10 replicates of Tn mutant libraries is quite an accurate method for detecting essential genes in nonstructural small genes such as small regulatory RNAs. However, as DeJesus shows, the number essential genes identified by TnSeq are comparable in large genes possessing more than 10 TA sites between 2 and 14 TnSeq datasets, most of which seem to be structural genes (Supplementary Fig 2 in mBio 2017; new Ref#57). Thus, we do not consider that we made a serious mistake for the classification of essentiality in most of the structural genes that encode proteins. With respect to the coverage of non-permissive sites, our TnSeq method might need to be improved if it is intended to classify the gene essentiality quite accurately on the small genes including small regulatory RNAs.

We investigated the non-permissive TA sites in ATCC13950. There are 4136 (6.43% of total ORFs) nonpermissive TA sites in ATCC13950, which is less than in H37Rv (9% of total ORFs) (DeJesus MA. mBio 2017; new Ref#57) and in *M. abscessus* ATCC19977 (8.1% of total ORFs) (Rifat D. mBio. 2021; new Ref#58). As for larger ORFs (TA sites ≥ 10), there are nonpermissive TA sites in 89 genes (ORFs) of common “essential (ES)” or “growth-defect-associated (GD)” (4.82% of a total of 1844 larger ORFs in ATCC13950). As for small ORFs (2-9 TA sites), there are nonpermissive TA sites in 41 genes (ORFs) of common ES or GD (1.35% of a total of 3021 smaller ORFs in ATCC13950).

We appreciate the idea of concluding the fate of gene essentiality by the presence/absence of non-permissive TA sites. However, we cannot conclude the fate of gene essentiality classification only by the presence/absence of potential non-permissive sites. Because, strictly to say, it is impossible to conclude the scientific truth of gene essentiality without functional analysis using gene manipulation. In accurate, TnSeq can “predict” the gene essentiality but cannot perfectly guarantee the functional significance. However, in the current situation, most of the recent TnSeq studies have been published only by the TnSeq analysis without functional analysis that uses gene manipulation strains of all targets they identified. Taking such limitations of TnSeq including non-permissive sites into consideration, we consider that the essentiality of the detected genes should be determined in further studies, mainly including biological experiments such as functional studies using gene manipulation strains.

We have added the above-mentioned contents in the revised manuscript (pages 32-33, lines 559-580).

[References]

Carey, A.F. et al. TnSeq of *Mycobacterium tuberculosis* clinical isolates reveals strain-specific antibiotic liabilities. PLoS Pathog 14, e1006939 (2018).

Minato, Y., et al. Genomewide assessment of *Mycobacterium tuberculosis* conditionally essential metabolic pathways. mSystems. 4, e00070-192019 (2019).

Akusobi. C. et al. Transposon-sequencing across multiple *Mycobacterium abscessus* isolates reveals significant functional genomic diversity among strains. mBio 6, e0337624 (2025).

DeJesus, M.A. et al. Comprehensive essentiality analysis of the *Mycobacterium tuberculosis* genome via saturating transposon mutagenesis. mBio 8, e02133-16 (2017).

Rifat, D., Chen L., Kreiswirth, B.N. & Nuermberger, E.L.. Genome-wide essentiality analysis of *Mycobacterium abscessus* by saturated transposon mutagenesis and deep sequencing. *mBio* 12, e0104921 (2021).

(1c) Line 100: Authors report a total of 131 genes identified as essential or growth-defect-associated with the HMM analysis across all *M. intracellulare* strains. It should be explained in more detail how gene essentiality was determined (see above comment in (1b)). Furthermore, in Table S3 authors should mention the essential and growth defective trait of each of the 131 genes.

Thank you for the comment on how to classify the 131 genes as essential or growth-defect-associated with the HMM analysis across all *M. intracellulare* strains. As replied in (1b), the average saturation of Tn insertion of our libraries became 62-79% when combining duplicate or triplicate data in each strain. The levels of saturation of transposon libraries in our study is similar to the very recent TnSeq analysis by Akusobi where 52-80% saturation libraries (so-called “high-density” transposon libraries) were used for HMM and resampling analyses, and most of triplicate libraries ranges 70-79% saturation (Supplemental Methods Table 1[merged saturation] in Akusobi. *mBio*. 2025; new Ref#9). The saturation of Tn insertion in individual replicates of our libraries is also comparable to those with DeJesus (Table S1 in *mBio* 2017; new Ref#57). Thus, we consider that our TnSeq libraries are acceptable for identifying essential genes and growth-defect-associated genes by the HMM method.

We used the HMM method as reported by DeJesus (DeJesus. *PLoS Comput Biol*. 2015; new Ref#12). HMM method can categorize the gene essentiality throughout the genome including “Essential”, “Growth-defect”, “Non-essential” and “Growth-advantage”. “Essential” genes are defined as no insertions in all or most of their TA sites. “Non-essential” genes are defined as regions that have usual read counts. “Growth-defect” genes are defined as regions that have unusually low read counts. “Growth-advantage” genes are defined as regions that have unusually high low read counts.

Following the previous report (Carey AF. *PLoS Pathog* 2018; new Ref#8), the annotation for the clinical MAC-PD strains was adapted from that of ATCC13950 by adjusting the START and END coordinates of each ORF in the clinical MAC-PD strains according to their alignment with the corresponding ORFs of ATCC13950. By using an adjusted annotation table, gene essentiality was classified by the HMM analysis.

We have added the explanation of how we identified essential and growth-defect-associated genes in the Methods (pages 35-36, lines 620-632). And following the comment, we have added the data of classification of gene essentiality in the 131 genes in the new Supplementary Table 3 in the revised manuscript.

[Reference]

DeJesus, M.A. et al. TRANSIT--A Software Tool for Himar1 TnSeq Analysis. *PLoS Comput Biol* 11, e1004401 (2015).

Carey, A.F. et al. TnSeq of *Mycobacterium tuberculosis* clinical isolates reveals strain-specific antibiotic liabilities. *PLoS Pathog* 14, e1006939 (2018).

Akusobi, C. et al. Transposon-sequencing across multiple *Mycobacterium abscessus* isolates reveals significant functional genomic diversity among strains. *mBio* 6, e0337624 (2025).

DeJesus, M.A. et al. Comprehensive essentiality analysis of the *Mycobacterium tuberculosis* genome via saturating transposon mutagenesis. *mBio* 8, e02133-16 (2017).

(1d) In Table S4, the authors show strain-specific putative essential genes from the core and accessory gene sets. For the sake of clarity, it is important to have the name of all the strains against each gene in which it is predicted essential or growth defective.

Thank you for the comment on the hit strains on the genes classified as strain-specific and accessory putative essential of growth-defect associated. Following the comment, we have added the data of hit strains in the new Supplementary Table 4 in the revised manuscript.

(1e) Lines 123-126: It is not clear what is the relevance of highlighting genes involved in hypoxic pellicle formation in ATCC13950. These appear to be randomly distributed across different clinical isolates and is not clear whether they correlate with differential susceptibility of the reference strain and clinical isolates to hypoxia.

Thank you for the comment on the relevance of highlighting genes involved in hypoxic pellicle formation in ATCC13950. The rationale for the importance of hypoxic pellicle genes in clinical MAC-PD strains is that the profiles of genetic requirements in each bacterial strain reflect the adaptation to the environment in which each strain lives. When the strains are placed in a special environment, they can adapt to the situation by altering the profiles of genetic requirements, resulting in the remodeling of metabolic pathways. We indeed found that the genetic requirements of several hypoxic pellicle genes were increased in clinical MAC-PD strains *in vitro* situations. These data suggest the hypoxic pellicle genes become more important in clinical MAC-PD strains for *in vitro* growth than in ATCC13950.

Moreover, hypoxia is known to be one of the characteristic conditions *in vivo* including clinical lesions (McKeown. Br Br J Radiol. 2014). We consider it reasonable to expect that the strains derived from MAC-PD patients without predisposing immunological disorders may adapt under hypoxic conditions for maintaining bacterial survival. Therefore, we highlighted the genes involved in hypoxic pellicle formation in ATCC13950.

We have added the description of the rationale for the importance of hypoxic pellicle genes in clinical MAC-PD strains in the revised manuscript (page 9, lines 148-155).

[Reference]

McKeown, S.R., et al. Defining normoxia, physoxia and hypoxia in tumours-implications for treatment response. Br Br J Radiol 87,; 20130676 (2014).

(2) Result 2 (pages 8-10): Genes with increased gene essentiality in clinical MAC-PD strains are also required for hypoxic pellicle formation in the type strain.

(2a) As reported by authors (lines 123-126), only a small fraction of genes showing essentiality in clinical MAC-PD strains are required for hypoxic pellicle formation in the reference strain, which might be due to random distribution. Authors should avoid making such a generalised statement that reflects the association of the entire essential gene pool in clinical MAC-PD strains with hypoxic pellicle formation.

Thank you for the comment on the issue of a small fraction of genes showing increased genetic requirements in clinical MAC-PD strains that is shared with genes required for hypoxic pellicle formation in the type strain ATCC13950. We admit that the section title may mislead that the genes required for hypoxic pellicle formation confer the entire essential gene pool of clinical MAC-PD strains. Following the comment, we have revised the section title as “Partial overlap of the genes showing increased genetic requirements in clinical MAC-PD strains with those required for hypoxic pellicle formation in ATCC13950” (page 9, lines 146-147).

We consider that it cannot be explained by a mere coincidence that we obtained the data of partial overlap of genes showing essentiality in clinical MAC-PD strains with genes required for hypoxic pellicle formation in ATCC13950, because we demonstrated the supporting data such as the pattern of genetic requirements suggesting gluconeogenic metabolic shift (Fig. 5) and the different pattern of hypoxic growth curves between clinical MAC-PD strains and ATCC13950 (Fig. 7).

(2b) I fail to understand how the number of Tn insertions determines "more" or "less" essentiality of a gene particularly with 50-60% saturation. To my understanding, essentiality is a qualitative trait. Either a gene will be essential (based on no Tn insertion despite having the permissive sites), critical (poor representation of Tn insertions at the permissive sites due to growth defect of the strain in the pool), non-essential (expected frequency of insertion) or growth-advantageous (higher representation of Tn insertions at the permissive sites due to growth advantage of the strain in the pool). Hence, authors should avoid quantifying the essentiality of a gene.

Thank you for the comments on the trait of gene essentiality. We realize that essentiality is a qualitative trait, not a quantitative trait. Taking into consideration the number of Tn insertions determines "more" or "less" requirements of a gene, we have corrected the manuscript by using the phrase "genetic requirements" instead of "gene essentiality".

As mentioned earlier, our method of comparison of genetic requirements between strains are the same as a previous report that used duplicate Tn mutant libraries of clinical Mtb strains of different genotypes and triplicate Tn mutant libraries of H37Rv for identifying increased genetic requirements of clinical Mtb strains (Carey AF. PLoS Pathog 2018; new Ref#8). Moreover, as described in rebuttal (1b), the saturation of our Tn mutant libraries by combining replicates are 62-79% as follows: ATCC13950: 67.6%, M001: 72.9%, M003: 63.0%, M018: 62.4%, M019: 74.5%, M.i.27: 76.6%, M.i.198: 68.0%, MOTT64: 77.6%, M021: 79.9%. That is, we calculated gene essentiality from the Tn mutant libraries with 62-79% saturation in each strain. The levels of saturation of transposon libraries in our study is similar to the recent TnSeq analysis by Akusobi where 52-80% saturation libraries ("high-density" transposon libraries) were used for HMM and resampling analyses (Supplemental Methods Table 1[merged saturation] in Akusobi C. mBio. 2025; new Ref#9).

Thus, we consider that our data of the difference of genetic requirements between clinical MAC-PD strains and ATCC13950 are acceptable.

[Reference]

Akusobi. C. et al. Transposon-sequencing across multiple *Mycobacterium abscessus* isolates reveals significant functional genomic diversity among strains. mBio 6, e0337624 (2025).

(2c) From Figures 3-4, it seems the authors intend to highlight the insertion frequencies of certain genes in the clinical isolates compared to those in the reference strain to conclude whether a gene has become more critical and its disruption results in the growth defective phenotype (poor representation) in the clinical isolates, or a critical/essential gene has become dispensable in these strains.

Based on these arguments, I suggest that the authors modify the title of the result such as "Tn insertion reveals differential requirement of genes for in vitro growth of clinical MAC-PD strains" or "Identification of genes differentially required for in vitro growth of clinical MAC-PD strains" as this is precisely the information we gain from this section of the study. Also, it is suggested to re-draft the rationale of this section as only 4 genes associated with hypoxic pellicle formation, were found to exhibit reduced insertion

frequencies in the clinical isolates out of total of 283 genes. Hypoxia-related genes can be highlighted in the next section (see below).

Thank you for the suggestion to modify the section title and to re-draft the rationale of the section. Following the comment, we modified the section title as “Partial overlap of the genes showing increased genetic requirements in clinical MAC-PD strains with those required for hypoxic pellicle formation in ATCC13950 (page 9, lines 146-147)

Following the suggestion, we have revised the rationale of this section as follows: “The sharing of strain-dependent and accessory essential or growth-defect-associated genes with genes required for hypoxic pellicle formation in ATCC13950 prompts us to consider that the profiles of gene essentiality in clinical MAC-PD strains may be associated with the genes required for hypoxic pellicle formation in ATCC13950.” (page 9, lines 151-155)

The reviewer points out that only 4 genes associated with hypoxic pellicle formation were found to exhibit reduced insertion frequencies in the clinical isolates out of total of 283 genes. However, to discuss how much proportion of the genes were detected to be increasingly required in clinical MAC-PD strains compared to ATCC13950, we should focus on the 121 genes showing increased requirements in clinical MAC-PD strains compared to ATCC13950, excluding the 162 genes indispensable for clinical MAC-PD strains. Thus, we described that 4 genes associated with hypoxic pellicle formation, were found to exhibit reduced insertion frequencies in the clinical isolates out of the 121 genes having significantly fewer Tn insertions than ATCC13950 in the manuscript (Fig. 3).

(3) Result 3 (Page 10-14): Requirement of genes with increased gene essentiality in the clinical MAC-PD strains for mouse lung infection.

(3a) The title should be modified to "Identification of genes in the clinical MAC-PD strains required for mouse lung infection".

Following the comment, we have modified the section title as "Identification of genes in the clinical MAC-PD strains required for mouse lung infection". (page 12, lines 201-202).

(3b) Further, the rationale of this experiment needs to be modified. As mentioned above, up until now the impact of hypoxic pellicle formation genes in the growth of MAC-PD strains remains unconvincing. The rationale of mouse infection experiments could be straightforward- "to identify genes critical for animal infection of the clinical isolates".

Thank you for the comment on the rationale of the *in vivo* TnSeq experiment. Following the comment, we have revised the rationale as “The impact of hypoxia on mycobacteria under various ecological circumstances implies that the genes required for pathogenesis of MAC-PD may be in some degrees, overlapped with the genes with increased requirements in the clinical MAC-PD strains compared to ATCC13950, and also with the genes required for hypoxic pellicle formation in ATCC13950. To identify genes required for *in vivo* infection of clinical MAC-PD strains,” in the revised manuscript (page 12, lines 204-210).

(3c) The authors should avoid using the term "genes with increased essentiality" for the reasons explained above in point #2b.

Following the comment, we have corrected the term as “genes with increased requirements” in the revised manuscript (page 12, line 207).

(3d) From Tables S8 and S9, I can find 93 genes in Mi198Tn and 74 genes in Mi27Tn for which Tn insertion mutants are under-represented in TnSeq at all time points from Day 1 to Wk 16 in comparison to input. Importantly, excluding results from Day 1 when the

infection has just settled, I find 172 and 121 genes in Mi198Tn and Mi27Tn, respectively, under-represented in lungs between Wk 4-16. My suggestion is that authors should focus more on such genes and identify the characteristics of these genes and what fraction belongs to those involved in hypoxic pellicle formation in the reference strain. I am perplexed why authors have categorically ignored other genes and only focused on a set of genes that correspond to ~10-12% of entire differentially abundant mutant pool.

Thank you for the suggestion on the genes that Tn insertion mutants are under-represented in TnSeq from Weeks 4-16 in the infected mouse lungs be analyzed for overlapping the genes involved in hypoxic pellicle formation in the type strain ATCC13950. We found that at all timepoints from Day1 to Week 16, 74 genes and 99 genes were under-represented in lungs infected with M.i.27Tn and M.i.198Tn, respectively. Of them, 21 (28.3%) and 12 (12.1%) genes belonged to the genes involved in the genes required for hypoxic pellicle formation in the type strain. We found that at timepoints from Week 4 to Week 16, 121 genes and 172 genes were under-represented in lungs infected with M.i.27Tn and M.i.198Tn, respectively. Of them, 21 (23.1%) and 30 (18.0%) genes belonged to genes involved in hypoxic pellicle formation in the type strain. These hypoxic pellicle-associated genes detected both in M.i.27 and M.i.198 encoded methionine synthesis, acyl-CoA dehydrogenase, isocitrate lyase, MMPL family transporter at all time points (from Day1 to Week 16). And additionally, multifunctional oxoglutarate decarboxylase/dehydrogenase, proteasome subunits, ABC transporter ATP-binding protein/permease, lipase chaperone at all time points (from Week 4 to Week 16). We have described these results in the Result section (page 14 lines 236-248) and new Supplementary Tables 12 and 13.

As for *M. intracellulare*, conditionally essential genes have not been revealed except for those required for hypoxic pellicle formation in ATCC13950 revealed by us (Tateishi Y. Sci Rep. 2020; new Ref#10). This study is the first to focus on the relationship between the difference of genetic requirements among strains and hypoxic adaptation. We found a certain proportion of overlapped genes required for mouse lung infection and ATCC13950's hypoxic pellicle formation. We consider it reasonable to focus on the category of genes required for hypoxic pellicle formation to analyze the datasets of TnSeq in mice.

[Reference]

Tateishi, Y. et al. Genome-wide identification of essential genes in *Mycobacterium intracellulare* by transposon sequencing - Implication for metabolic remodeling. Sci Rep 10, 5449 (2020).

(3e) Page 13, lines 224-227: "Despite the differences in the profiles of the genes required for in vivo infection between strains, these data suggest that increased gene essentiality for hypoxic growth confers advantages for pathogenesis in vivo."

For the reason described above, I find it a misleading hypothesis that hypoxic growth confers advantages for pathogenesis in vivo. How come only 10-12% of the entire gene sets which include genes of varying functions, can be the sole contributors to bacterial survival in host organelles during infection?

More importantly, the mouse is not considered a good model for hypoxia as mouse infection does not lead to the formation of solid granuloma with a hypoxic core Though I am not convinced with the authors' bias toward hypoxia-related genes, however, if at all they aim to investigate the role of such genes by an unbiased enrichment of TnSeq mutant, they should have used C3HeJ mice which are known to form granulomas (Boute et al., 2017 (doi: 10.1186/s13567-017-0477-7)).

Thank you for the comments on the issue of the contribution of genes required for hypoxic growth and on the difference of hypoxic levels between mouse lineages. We did not intend to

mention that a set of genes required for hypoxic growth is the sole contributor to bacterial survival in host organs during infection. As we discussed in the Discussion section, we acknowledge that the adaptation to the difference of carbon source between *in vitro* and *in vivo* infection (i.e. preferential usage of lipid carbon source *in vivo*) is involved in the pathogenesis of mycobacterial diseases (Yang. Front Microbiol 2018; new Ref#33, Gouzy. Proc Natl Acad Sci U S A 2021; new Ref#29, Quinonez. mBio 2022; new Ref#40, Pandey. Proc Natl Acad Sci U S A 2008; new Ref#41). We consider that not only the genes required for hypoxic pellicle formation but also strain-dependent/accessory genes conferring kinds of metabolism other than hypoxic pellicle formation can be estimated to be involved in the *in vivo* mouse lung infection.

We have modified the sentence to clearly express our intention as follows: “These *in vivo* TnSeq data suggest that, despite the differences in the profiles of the genes required for *in vivo* infection between strains, increase of genetic requirements for hypoxic growth in part contribute to the pathogenesis *in vivo*.” (pages 15-16, lines 269-271)

It seems to be an interesting idea to perform TnSeq by using C3HeJ mice. The granuloma formed in C3HeJ mice becomes extremely hypoxic (less than 1%, corresponding the level of “pathological” hypoxia) which is as severe as the detection range by pimonidazole. In our model, the effect of such pathological levels of hypoxia on granuloma formation might not be detected. However, the lesion formed in C57BL/6 mice becomes a “physiological” level of hypoxia (5% O₂) (McKeown SR. Br Br J Radiol. 2014) which is the same O₂ level for *M. intracellulare* to form pellicles. In principle, oxygen levels inside human bodies are physiologically hypoxic, and many biological events are experimentally investigated in this condition. Thus, we consider that we were able to observe the effect of physiological hypoxia on *M. intracellulare* growth both *in vitro* (hypoxic pellicles) and *in vivo* (infected C57BL/6 mice).

[Reference]

Yang, T. et al. Pan-genomic study of *Mycobacterium tuberculosis* reflecting the primary/secondary genes, generality/individuality, and the interconversion through copy number variations. Front Microbiol 9, 1886 (2018).

Gouzy, A., Healy, C., Black, K.A., Rhee, K.Y. & Ehrt, S. Growth of *Mycobacterium tuberculosis* at acidic pH depends on lipid assimilation and is accompanied by reduced GAPDH activity. Proc Natl Acad Sci U S A 118, e2024571118 (2021).

Quinonez, C.G. et al. The role of fatty acid metabolism in drug tolerance of *Mycobacterium tuberculosis*. mBio 13, e0355921 (2022).

Pandey, A.K. & Sassetti, C.M. Mycobacterial persistence requires the utilization of host cholesterol. Proc Natl Acad Sci U S A 105, 4376-4380 (2008).

McKeown, S.R. et al. Defining normoxia, physoxia and hypoxia in tumours-implications for treatment response. Br Br J Radiol 87, 20130676 (2014).

(3f) An important set of data with the ATCC13950 reference strain is missing here. It is suggested that authors perform this study with the reference strain to identify whether the enrichment of genes is similar across all strains or is specific to the clinical isolates.

Thank you for the comment on the setting of ATCC13950 as a control strain in the mouse infection experiment. However, we proved that bacterial burden of ATCC13950 is reduced continuously from 4 weeks of infection, and that ATCC13950 is almost completely eliminated from 8 to 16 weeks of infection (BMC Microbiol 2023; new Ref#22). Therefore, it is impossible to perform TnSeq to detect the genes required for persistent infection in mice infected with ATCC13950.

[Reference]

Tateishi, Y. et al. Virulence of *Mycobacterium intracellulare* clinical strains in a mouse model of lung infection - role of neutrophilic inflammation in disease severity. BMC Microbiol 23, 94 (2023).

(3g) Pages 13-14, lines 228-245: "We have performed a statistical enrichment analysis of gene sets by GSEA..."

The comparison made here is not clear to me. It seems the authors do compare genes required for the growth of M.i.27 and M.i.198 in mouse lungs with the gene sets required for hypoxic pellicle formation in ATCC13950 together with the gene sets showing increased gene essentiality observed in the clinical MAC-PD strains, and claim that a significant % of genes belong to hypoxia-adaptation pathways. It is factually incorrect because a majority of these might overlap with those found critical for the in vitro survival of MAC-PD strains. It is suggested that authors re-analyze their data by comparing genes required for the growth of M.i.27 and M.i.198 in mouse lungs individually with those involved in hypoxic pellicle formation in ATCC13950, and with the gene sets found critical for in vitro growth, and present accordingly.

Thank you for the suggestion on the re-analysis of gene enrichment analysis of genes required for M.i.27 and M.i.198 *in vivo* infection, individually with genes involved in hypoxic pellicle formation in ATCC13950 and with those showing genetic requirements in clinical MAC-PD strains compared to ATCC13950.

About 50% (92 and 94 out of 181 genes through Day 1 to Week 16 and through Week4 to Week16 of infection) and 40% (70 and 79 out of 179 genes through Day 1 to Week 16 and through Week 4 to Week 16 of infection) of genes required for hypoxic pellicle formation in ATCC13950 were listed as enriched in genes required for mouse lung infection in M.i.27 and M.i.198, respectively. In addition, about 42% (54 and 56 out of 128 genes through Day 1 to Week 16 and through Week 4 to Week 16 of infection) and 40% (79 and 68 out of 179 genes through Day 1 to Week 16 and through Week 4 to Week 16 of infection) of genes showing increased requirements in clinical MAC-PD strains compared to ATCC13950 were listed as enriched in genes required for mouse lung infection in M.i.27 and M.i.198, respectively.

The tables and graphs of GSEA results are shown in Supplementary Figs. 5, 6.

These data indicate that 40-50% of the genes required for *in vitro* hypoxic pellicle formation and the strain-dependent/accessory essential genes are significantly enriched individually with *in vivo* bacterial growth. We have added the result of reanalyzed data of GSEA in the Result (pages 16-17, lines 287-290). We have shown the detail of reanalyzed data of GSEA in Supplementary Figs. 5, 6 and Supplementary Tables 15, 16.

(3h) *Since authors have used Tnseq of pooled mutants, which often yields misleading information, it is important to validate some of their findings upon mouse infection with individual mutants that yield prominent as well as baseline reduction at different time points. In the absence of validation, it remains a mere speculation of the role of these genes in the infection of these strains to animals.*

Thank you for the suggestion on the validation of the TnSeq hit genes on the *in vivo* survival. We acknowledge the importance of validating the TnSeq-hit genes by constructing knockout mutants. We have recently succeeded in constructing the vectors for making knockout strains of *M. intracellulare* (Tateishi Y. Microbiol Immunol. 2024). We will proceed to the infection experiment of knockout mutants by using our system for constructing them.

[Reference]

Tateishi Y. et al. Construction of knockout mutants in *Mycobacterium intracellulare* ATCC13950 strain using a thermosensitive plasmid containing negative selection marker *rpsL*+. Microbiol Immunol 68, 339-347 (2024).

(4) Result 4 (Page 14-15): Preferential hypoxic adaptation of clinical MAC-PD strains evaluated with bacterial growth kinetics.

(4a) "The metabolic remodeling, such as the increased gene essentiality of gluconeogenesis and the type VII secretion system..". As stated above, the essentiality of a gene, being a qualitative trait, should not be presented in quantitative terms. The authors should re-phrase this statement.

Following the comment, we have corrected the term as “The metabolic remodeling, such as the increased genetic requirements of gluconeogenesis and the type VII secretion system.” (page 17, lines 296-297)

(4b) "overlap of the genes required for mouse lung infection and those required for hypoxic pellicle formation involved by conferring these metabolic pathways..". There is a syntax error in this statement and needs revision.

Following the comment, we have corrected the phrase as “overlap of the genes required for mouse lung infection and those required for hypoxic pellicle formation involved by these metabolic pathways”. (page 17, lines 297-299)

(4c) The altered requirement of genes in different clinical strains for survival provides only circumstantial evidence of metabolic remodeling. Authors are suggested to perform metabolic profiling of representative clinical and reference strains, as it is important to examine whether these bacteria indeed undergo metabolic shift.

Thank you for the comment on the metabolic profiling of the representative clinical and reference strains. We previously published the TnSeq result of ATCC13950 and we produced the current data by organizing with our previous findings (Fig. 4 in Tateishi Y. Sci Rep 2020; new Ref#10). The priority of the current study was to elucidate the difference and diversity of genetic requirements between clinical MAC-PD strains and ATCC13950. We consider that it is of some value to show the even circumstantial evidence of metabolic remodeling by TnSeq, because it provides a strong rationale for proceeding to the next study including metabolomic analysis.

[Reference]

Tateishi, Y. et al. Genome-wide identification of essential genes in *Mycobacterium intracellulare* by transposon sequencing - Implication for metabolic remodeling. Sci Rep 10, 5449 (2020).

(5) Result 5 (Page 16-18): Effects of knockdown of universal and accessory/strain-dependent essential or growth-defect-associate genes in clinical MAC-PD strains.

(5a) Lines 273-277: The rationale of using CRISPRi should be correctly presented to evaluate the effect of individual genes' suppression on the downstream phenotype and not to establish the CRISPRi silencing tool in MAC.

Thank you for the comment on the rationale of the section of the CRISPR-i experiment. Following the comment, we have modified the sentence as follows: “With an intention to

evaluate the effect of suppressing TnSeq-hit genes on bacterial growth.” (page 19, lines 333-334 in the revised manuscript).

(5b) Line 278: *pRH2052/pRH2521 are the plasmids and not the CRISPRi system.*

Following the comment, we have corrected the phrase as “pRH2052/pRH2521 clustered regularly interspaced short palindromic repeats interference (CRISPR-i) plasmids.” (page 19, lines 334-335 in the revised manuscript).

(5c) Line 280: *Other pioneering studies on the use of CRISPRi for gene silencing in mycobacteria (Chaudhary et al., Nat Comm, Rock et al., Nat Microbio) should also be cited.*

Thank you for the comment on adding the reference papers on CRISPR-i in mycobacteria. We have added the two suggested papers in the revised manuscript as new Ref #30 and #31. (page 19, line 336).

(5d) Lines 282-283: *It is not clear why M001 and MOTT64 could not be transformed. Did the authors use any control plasmid to evaluate the transformation efficiency of these strains?*

Thank you for the comment on the failure of transformation in M001 and MOTT64.

Following the comment, we have performed the experiment for evaluating the efficiency of transformation in the 9 *M. intracellulare* strains we used in this study. We have used an *E. coli*-mycobacteria shuttle vector pSO246KM-Prhsp65-luc that expresses firefly luciferase as a control plasmid (Aoki K. J Biol Chem 2004). For obtaining transformed colonies, we used 7H10/OADC agar plates containing the same concentration of kanamycin that we used for preparing Tn mutant libraries and for obtaining CRISPR-i knockdown strains.

We have observed no colonies grown on agar plates in MOTT64 after electroporation of the pSO246KM-Prhsp65-luc plasmid. In most of the remaining strains, the transformed colonies have emerged fully on day 10 of culture after electroporation of the plasmid. However, we have observed that M001 needs twice as long as a period for the emergence of transformed colonies. On day 21, the number of colonies in M 001 have finally become comparable to that of the other strains. We have checked the luciferase activity of 6-12 colonies in each strain except for MOTT64, and we have confirmed the transformation of the plasmid by the data of higher luciferase activity in the colonies undergoing electroporation of the plasmid than in those not undergoing electroporation.

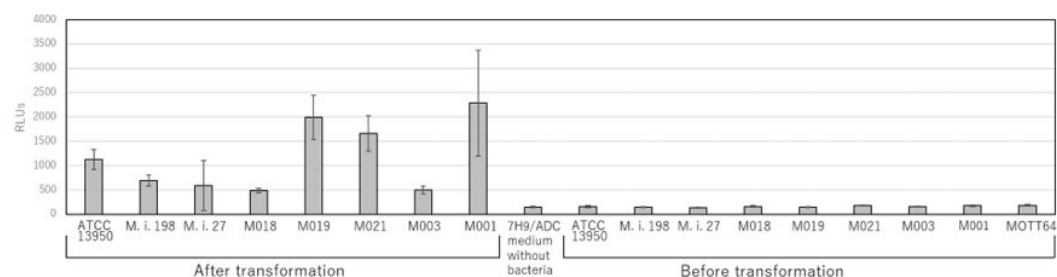
The possible reason for the incapability of obtaining transformants of CRISPR-i vectors in MOTT64 may be due to the extremely low efficiency of acquiring foreign DNA. And the possible reason for the incapability of obtaining transformants of CRISPR-i vectors in M001 may be intolerable to the stress caused by transformation of plasmids compared to other *M. intracellulare* strains. For M001, pSO246KM-Prhsp65-luc plasmid may cause tolerable stress for transformation, resulting in the delayed emergence of transformed colonies. By contrast, the CRISPR-i plasmids may cause greater stress for M001 than pSO246KM-Prhsp65-luc plasmid, resulting in being intolerable for transformation.

Author response table 1.

Strain	Transformation efficiency Day 10	Transformation efficiency Day 21
ATCC13950	8.24×10^{-6}	9.51×10^{-6}
M019	9.01×10^{-4}	9.01×10^{-4}
M021	1.93×10^{-3}	1.93×10^{-3}
M001	1.07×10^{-8}	6.32×10^{-6}
MOTT64	0.00	0.00
M.i.198	2.57×10^{-4}	2.57×10^{-4}
M.i.27	1.77×10^{-4}	4.17×10^{-4}
M018	4.10×10^{-4}	4.10×10^{-4}
M003	7.62×10^{-5}	7.62×10^{-5}

Author response image 3.

Result of luciferase activities before and after transformation of pS0246KM-Prhsp65-luc plasmid. Fifty microliter of cultures were mixed with 50 u L of assay reagents (Luciferase assay system E1500, Promega) and luciferase activity was measured by the luminometer (FilterMax F5, Molecular Devices). Data are shown as mean $\pm$ SD of 6-12 colonies



[Reference]

Aoki K. Extracellular mycobacterial DNA-binding protein 1 participates in *Mycobacterium*-lung epithelial cell interaction through hyaluronic acid. J Biol Chem 279, 39798–39806 (2004).

(5e) Lines 283-186: "To confirm the gene essentiality detected with the HMM analysis, we evaluated the consequent growth inhibition in the knockdown strains of representative universal essential or growth-defect-associated genes, including *glcB*, *inhA*, *gyrB*, and *embB*." It is not clear what was the level of suppression of these genes in the respective KD strains. Authors should include the level of suppression of these genes also by qRT-PCR.

Thank you for the comment on the suppression levels of gene expression in knockdown strains of universal essential genes. Following the comment, we have evaluated them by qRT-PCR and we observed comparable levels of knockdown efficiency in the knockdown strains between universally essential genes and strain-specific/accessory essential genes (new

Supplementary Fig. 9). Overall, the gene expression was suppressed to 20 - 70% in the knockdown strains compared to the vector control strains that do not express sgRNA.

We have added the data of qRT-PCR of knockdown strains of universal essential genes such as *glcB*, *inhA*, *gyrB*, and *embB* (new Supplementary Fig. 9). We have revised the Result and Discussion in the manuscript (page 21, lines 367-376; page28, lines 490-497).

(5f) Lines 293-: I am unable to establish any correlation between the growth of the knockdown with Tn insertion reads in the respective genes. For instance, *pckA* exhibits reduced Tn insertion reads in almost all the strains except in M.i.27, but the effect of its KD on growth is seen only in M.i.198 and M003; *glpX* exhibits reduced Tn insertion reads in M003, M019, M021 but the effect of its KD on growth is seen only in M003; *csd* exhibits reduced Tn insertion reads in M.i.198, M003, M019 but the effect of its KD on growth is seen only in M.i.198 and M003. The authors argue that these contradictory phenotypes are due to difficulties in the effective operation of genetically modified systems using foreign genes from different bacterial species in MAC-PD strains (Lines 310-312) or the desired effect on growth could not be observed due to the inability of CRISPRi to yield >99% suppression (Line 314) are not the valid justifications. Indeed, a close look at the RT-PCR data (Figure S5) reveals that *pckA* levels are ~0.22, 0.5, 0.2, 0.22, 0.2, 0.5, and 0.3 fold relative to *sigA* in M.i.198, M.i.27, ATCC13950, M018, M019, M003 and M021, respectively, but the effect of its suppression on growth by CRISPRi is seen only in M.i.198 and M003. Secondly, >99% suppression is not a universal prerequisite for all the genes to show growth defect (as might be the case with *glcB*, *inhA*, *gyrB*, and *embB* genes in this study). Hence, it remains unclear why contrasting results are obtained for most of the genes by TnSeq and CRISPRi.

Thank you for the comments on the issue of inconsistent results between TnSeq and CRISPR-i based knockdown. We acknowledge that some inconsistencies were observed, particularly among strain-dependent/accessory essential or growth-defect associated genes. By contrast, we found consistent data between TnSeq and CRISPR-i based knockdown results of universal essential genes. By obtaining the data of suppression levels of gene expression in the knockdown strains of universal essential genes, we have acknowledged that the low efficiency of knockdown does not explain the reason of the discrepancy between TnSeq and CRISPR-i results because the levels of knockdown efficiency were comparable between strain-dependent/accessory essential genes and universally essential genes.

Although the mechanism has not been fully proven yet only from the current study, we consider that such inconsistent phenotypes with TnSeq and CRISPR-i based knockdown may be related to the recently revealed the bypass mechanism of gene essentiality which is characteristically observed in strain-dependent/accessory essential or growth-defect-associated genes. According to the publication by Rosconi (Nat Microbiol. 2022: new Ref#14) reporting the ‘forced-evolution experiments’ of 36 clinical *Streptococcus pneumoniae* strains, gene essentiality can be bypassed by several mechanisms including the composition of the accessory genome and pathway rewiring. They recovered successfully knockout mutants from transformation experiments in strain-specific/accessory essential genes such as cytidine monophosphate kinase, a folate pathway enzyme formate tetrahydrofolate ligase and an undecaprenyl phosphate-biosynthesis pathway enzyme farnesyl-diphosphate synthase. The bypassing of gene essentiality could be suggested by observing suppressor mutations and synthetic lethality in knockout strains. By contrast, universal essential genes were reported to fulfill the three categories including high levels of conservation within and often across species, limited genetic diversity, and high and stable expression levels. Consequently, universal essential genes are estimated to be rigid, largely immutable key components to an organism’s survival.

We consider that this is the case with our study on NTM because NTM is pangenomic. The knockdown of universal essential genes resulted in the clear growth suppression; however, the knockdown of strain-dependent/accessory essential genes did not show the consistent growth suppression. We consider that the bypass mechanism of gene essentiality can explain the inconsistent effect of gene silencing of strain-dependent/accessory genes on bacterial growth suppression.

We have added the above-mentioned description in the Discussion (pages 28-29, lines 497-519).

[Reference]

Rosconi, F. et al. A bacterial pan-genome makes gene essentiality strain-dependent and evolvable. *Nat Microbiol* 7, 1580–1592 (2022).

Minor Comments:

(1) The authors should mention the cut-off of fold-change for all the experiments in the methods section.

Thank you for the comment on the cut-off of fold-change. We set the cut-off of fold-change as adjusted P -value < 0.05 . We added the description in the Methods section. (page 41, lines 724-725)

(2) Figure 7 legend (Lines 888-889): "Data are shown as the means {plus minus} SD of triplicate experiments. Data from one experiment representative of three independent experiments (N = 3) are shown."

Figure S3 legend: Data on the growth curves are the means of triplicate experiments. Data from one experiment representative of three independent experiments (N = 3) are shown.

Figure S4 legend: Data are shown as the means {plus minus} SD of triplicate experiments. Data from one experiment representative of two independent experiments (N = 2) are shown.

Figure S5 legend: Gene expression data are the means {plus minus} SD of triplicate experiments. Data from one experiment representative of two independent experiments (N = 2) are shown.

These statements need clarification. Whether multiple independent experiments (biological repeats), each with 2-3 technical replicates performed and the data shown represent one of the multiple biological repeats?

Thank you for the comments on the number of experiments performed and the number of replicates. We have performed two or three independent experiments with 2-3 technical replicates. The data shown represent one of the independent experiments.

(3) Figure 7b: Statistics are missing in the bar graph for growth rate under aerobic conditions.

Thank you for the comment on the statistics of the data regarding growth rate under aerobic conditions. We have added the statistics in the new Fig. 7c.

(4) The authors should check the y-axis in Figure 7b, as it is not clear whether bacteria indeed show a growth rate of 1-3 CFUs/day.

Thank you for the comment on the y-axis in Figure 7b. We have corrected the label of y-axis as “log₁₀[CFUs]/day” in the new Fig. 7c. Additionally, we have corrected the label of y-axis in new Fig. 7a and added the description as “Data are represented as CFUs in 4 µl sample at each timepoint.” in the Fig. 7a legend.

Reviewer #3 (Recommendations For The Authors):

(1) It's notable that strains M001 and MOTT64 failed to undergo a transformation, while seven other strains did. Given that M001, MOTT64, and M019 belong to the same phylogenetic clade, it raises questions about why particular strains within this clade showed different transformation outcomes. It might be valuable for them to discuss this discrepancy in their study.

Thank you for the comment on the difference in capacity of transformation between strains belonging to the same genomic subgroup. Although the direct mechanism determining the competency for foreign DNA has not been elucidated in *M. intracellulare* and other pathogenic NTM species, several studies on general bacteria suggest the difficulties of introducing foreign DNA into clinical strains compared to the laboratory strains. As suggested in *Staphylococcus aureus* (Covaglia AR. PNAS. 2010; new ref#55), some clinical strains develop elimination system of foreign nucleic acids such as a type III-like restriction restriction endonuclease. As suggested in gram-negative bacteria (Qin J. Sci Rep. 2022; new Ref#56), there may be some difference in cell surface structures between strains, resulting in the necessity of polymyxin B nonapeptide targeting cell membrane for transforming clinical strains. The efficiency of eliminating foreign DNA may be attributed to various kinds of strain-specific factors including restriction endonuclease, natural CRISPR-interference system and cell wall structures rather than a simple genotypic factor.

We have added the description on the difference of capability in transformation in the Discussion. (page 31, lines 546-558)

[References]

Covaglia, A.R., François, P., Hernandez, D., Perron, K., Linder, P. & Schrenzel, J. A type III-like restriction endonuclease functions as a major barrier to horizontal gene transfer in clinical *Staphylococcus aureus* strains. Proc Natl Acad Sci U S A 107, 11954-11958 (2010).

Qin, J., Hong, Y., Pullela, K., Morona, R., Henderson, I.R. & Totsika, M. A method for increasing electroporation competence of Gram-negative clinical isolates by polymyxin B nonapeptide. Sci Rep 12, 11629 (2022).

(2) The authors should consider specifying *M. intracellulare* in their title.

Thank you for the comment on the manuscript title. Following the comments from all Reviewers, we have modified the title as “Functional genomics reveals strain-specific genetic requirements conferring hypoxic growth in *Mycobacterium intracellulare*”.

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